

DFlowNovo: Continuous-Time Discrete Flow Matching and Dynamic Knapsack Guidance for High-Throughput De Novo Peptide Sequencing

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September 2026

Submitted in partial fulfillment of a structured master's degree at AIMS South Africa



Abstract

Tandem mass spectrometry (MS/MS) enables high-throughput identification of proteins, yet identifying novel peptides without reference databases—*de novo* peptide sequencing—remains computationally challenging. Existing state-of-the-art approaches rely on autoregressive transformers that predict amino acid sequences token by token in $\mathcal{O}(L)$ sequential steps. While accurate, autoregressive decoding suffers from error propagation, directional fragmentation asymmetry between complementary *b*- and *y*-ions, and severe latency bottlenecks. Continuous normalizing flows alleviate sequential decoding but suffer from severe discretization error upon rounding continuous latent states to discrete amino acid masses.

In this work, we introduce **DFlowNovo**, a non-autoregressive generative framework that models *de novo* peptide sequencing as continuous-time Discrete Flow Matching (DFM) over the probability simplex of amino acids, coupled with exact Dynamic Programming Knapsack reachability filtering (KnapsackDP). DFlowNovo generates complete peptide sequences in a constant budget of $K = 20$ reverse flow steps, achieving an inference throughput of 227.1–255.8 spectra per second on an NVIDIA GPU (a $4.3\times$ to $4.9\times$ speedup over autoregressive beam search). On the standardized 104k Nine-Species test benchmark, DFlowNovo achieves 68.28% strict exact match and 83.01% amino acid F1 score, outperforming InstaNovo v1.2 (65.48%) and accepting 86,412 peptide-spectrum matches at 80% precision. Furthermore, we dissect the physical foundations of mass spectrometry, demonstrating that remaining prediction discrepancies on human data are predominantly driven by isomeric Leucine/Isoleucine equivalence (113.084 Da) which requires radical-driven *w*-ion fragmentation to break.

Résumé

La spectrométrie de masse en tandem (MS/MS) permet l'identification des protéines à haut débit, mais le séquençage *de novo*—l'identification des peptides sans base de données de référence—reste un défi informatique majeur. Les modèles autorégressifs actuels prédisent les acides aminés de manière séquentielle en $\mathcal{O}(L)$ étapes, ce qui induit une propagation des erreurs, une asymétrie entre ions complémentaires *b* et *y*, et une forte latence. Les modèles à flux continus évitent ce coût séquentiel mais souffrent d'une perte de discrétisation critique lors de la projection sur les masses discrètes des résidus.

Ce travail présente **DFlowNovo**, un cadre génératif non-autorégressif formulant le séquençage *de novo* comme un appariement de flux discret (DFM) en temps continu sur le simplexe de probabilité, guidé par une programmation dynamique exacte du sac à dos (KnapsackDP). DFlowNovo décode les séquences peptidiques entières en un nombre fixe de 20 pas d'intégration, atteignant une vitesse de 227 à 256 spectres par seconde (4,3 à 4,9 fois plus rapide que les modèles autorégressifs). Sur le banc d'essai biologique des Neuf Espèces (104 163 spectres), DFlowNovo atteint 68,28% d'exactitude stricte et 83,01% de score F1, surpassant InstaNovo v1.2 tout en identifiant 86 412 spectres à 80% de précision. Enfin, une analyse biophysique démontre que les ambiguïtés résiduelles découlent de l'isomérisation fondamentale Leucine/Isoleucine, indissociable par fragmentation collisionnelle standard.

Declaration

I, the undersigned, hereby declare that the work contained in this research project is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.

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Joël Gédéon, September 2026

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1. Introduction to De Novo Peptide Sequencing and Mass Spectrometry

Proteins are the primary functional macromolecular effectors of biological processes, executing enzymatic catalysis, structural scaffolding, signal transduction, and immunological defense within all living organisms. Deciphering the exact amino acid sequence of proteins and peptides is therefore foundational to molecular biology, pharmacology, clinical oncology, and biotechnology. Tandem mass spectrometry (MS/MS) coupled with liquid chromatography (LC-MS/MS) has emerged as the premier high-throughput analytical technology for resolving complex cellular proteomes. However, translating high-dimensional, noisy fragmentation spectra into accurate primary sequences remains one of the most demanding inverse problems in computational biology.

1.1 The Proteomics Frontier and Tandem Mass Spectrometry

In a standard bottom-up proteomics workflow, extracted proteins are enzymatically digested into shorter peptide fragments (typically 7 to 35 amino acids in length) using endopeptidases such as trypsin (Olsen et al., 2004). These peptides are separated by reversed-phase high-performance liquid chromatography and introduced into a mass spectrometer via electrospray ionization (ESI).

During the initial survey stage (MS1), the instrument measures the mass-to-charge ratios (m/z) and isotopic abundance envelopes of intact, multiply protonated peptide precursor ions. Subsequently, selected precursor ions within narrow isolation windows are accelerated into a collision cell for tandem mass spectrometry (MS2). In Higher-Energy Collisional Dissociation (HCD) (Olsen et al., 2007), precursor ions collide with neutral inert gas molecules (such as nitrogen or argon), converting kinetic energy into vibrational excitation. This vibrational excitation induces unimolecular thermal decomposition across the weakest covalent bonds of the peptide backbone, predominantly the amide peptide bonds (Paizs and Suhai, 2005). The resulting fragment ions are injected into an ultra-high-resolution mass analyzer, such as an Orbitrap, which records their exact m/z and intensity values, generating an experimental MS2 fragmentation spectrum.

1.2 Database Search Engines versus De Novo Sequencing

For over two decades, the dominant paradigm for analyzing tandem mass spectra has been database-dependent search algorithms, pioneered by SEQUEST (Eng et al., 1994) and subsequently refined in tools such as Mascot, MaxQuant, and MSFragger. Database search engines operate by matching experimental spectra against *in silico* theoretical spectra generated from reference proteome sequence databases (such as UniProt). While highly sensitive and computationally tractable for well-characterized model organisms, database searching possesses severe intrinsic limitations (Nesvizhskii, 2014):

- (i) **Unannotated Genomes and Non-Model Species:** For the vast majority of biological species on Earth, high-quality reference genomes or proteome annotations do not exist. Database searching fails entirely when the target organism is unsequenced.
- (ii) **Cancer Neoantigens and Somatic Mutations:** Malignant cells generate novel peptide sequences resulting from point mutations, gene fusions, alternative splicing, and frame-shift events that are absent from canonical reference databases. Identifying these tumor-specific antigens is vital for personalized immunotherapy and cancer vaccine development.

- (iii) **Immunopeptidomics and Antibody Repertoires:** B-cell and T-cell receptor repertoires undergo somatic hypermutation and $V(D)J$ genetic recombination, generating hypervariable complementarity-determining regions (CDRs). Assembling monoclonal antibodies or polyclonal serum repertoires directly from biological fluids requires sequence identification without genomic references.
- (iv) **Post-Translational Modifications (PTMs):** Variable chemical modifications (phosphorylation, acetylation, ubiquitination, methylation) expand the combinatorial search space exponentially, rendering brute-force database searches computationally intractable.

To overcome these barriers, *de novo* peptide sequencing aims to reconstruct the exact primary amino acid sequence directly from the experimental spectrum $\mathcal{S}$ and precursor mass M_{prec} , relying solely on the physical laws of mass conservation and fragmentation chemistry.

1.3 Physical Principles of Collisional Fragmentation

To understand the mathematical structure of the sequencing problem, we examine the nomenclature and mechanisms governing peptide fragmentation. Following the standardized Roepstorff-Biemann nomenclature (Roepstorff and Fohlman, 1984; Biemann, 1988), cleavage of the amide bond ($-CO-NH-$) along the peptide backbone generates two complementary series of fragment ions:

1.3.1 Definition (Prefix and Suffix Fragment Ions). Let $Y = (y_1, y_2, \dots, y_L)$ denote a peptide of length L , where each $y_i \in \mathcal{V}_{\text{AA}}$ is an amino acid residue with monoisotopic mass $m(y_i)$.

- (i) A ***b*-ion** of length $k \in \{1, \dots, L-1\}$ contains the N-terminal prefix sequence $(y_1, \dots, y_k)$. Its theoretical monoisotopic mass is:

$$m(b_k) = \sum_{j=1}^k m(y_j) + m(\text{H}^+) \quad (1.3.1)$$

where $m(\text{H}^+) \approx 1.007276$ Da is the proton mass.

- (ii) A ***y*-ion** of length $k \in \{1, \dots, L-1\}$ contains the C-terminal suffix sequence $(y_{L-k+1}, \dots, y_L)$. Its theoretical monoisotopic mass is:

$$m(y_k) = \sum_{j=L-k+1}^L m(y_j) + m(\text{H}_2\text{O}) + m(\text{H}^+) \quad (1.3.2)$$

where $m(\text{H}_2\text{O}) \approx 18.010565$ Da represents the terminal water molecule retained by the carboxylic acid terminus.

Because cleavage of a specific peptide bond produces either a *b*-ion or a *y*-ion, every b_k ion has a complementary y_{L-k} ion satisfying the fundamental mass conservation duality:

$$m(b_k) + m(y_{L-k}) = M_{\text{prec}} + m(\text{H}^+) \quad (1.3.3)$$

where $M_{\text{prec}} = \sum_{i=1}^L m(y_i) + m(\text{H}_2\text{O}) + m(\text{H}^+)$ is the singly protonated intact precursor mass. The mass difference between two consecutive fragment ions of the same series reveals the mass of the intervening amino acid:

$$m(b_k) - m(b_{k-1}) = m(y_{L-k+1}) - m(y_{L-k}) = m(y_k) \quad (1.3.4)$$

In an idealized, noise-free mass spectrum, a complete ladder of *b*-ions or *y*-ions would permit trivial sequence extraction by reading off adjacent mass intervals. However, real experimental spectra deviate

profoundly from this idealization due to missing fragmentation peaks, internal fragments, neutral losses (H_2O , NH_3), multiply charged ions, chemical noise, and instrument measurement jitter (Paizs and Suhai, 2005).

1.4 Computational Complexity and Predecessor Algorithms

The primary amino acid alphabet consists of 20 canonical residues with monoisotopic masses ranging from Glycine (57.021 Da) to Tryptophan (186.079 Da). For a typical tryptic peptide of length $L \approx 15$, the number of possible sequence permutations is $20^{15} \approx 3.28 \times 10^{19}$. Filtering by precursor neutral mass restricts the candidate space, but still leaves millions of isobaric sequence permutations.

Early computational approaches framed *de novo* sequencing as finding the highest-scoring path through a directed acyclic spectrum graph (Dancik et al., 1999; Frank and Pevzner, 2005). Nodes represented observed experimental peaks, and directed edges corresponded to valid amino acid residue masses. While dynamic programming could compute optimal paths in polynomial time, these heuristic graph methods struggled with missing peaks, false ladder connections, and non-linear peak intensity variations.

The advent of deep learning revolutionized the field. DeepNovo (Tran et al., 2017) combined convolutional neural networks (CNNs) for peak feature extraction with long short-term memory (LSTM) networks and beam search decoding. PointNovo (Qiao et al., 2021) introduced continuous order-invariant spectrum encodings. Casanovo (Yilmaz et al., 2022) adapted the Transformer architecture, framing sequencing as an autoregressive machine translation problem from peak coordinates to amino acid tokens. More recently, InstaNovo (Eloff et al., 2025) incorporated mass-guided beam search with deep autoregressive Transformers, setting a new benchmark for sequencing accuracy across large-scale human proteomes.

Despite their success, existing autoregressive models suffer from fundamental algorithmic limitations:

- (i) **Sequential Decoding Latency:** Generating a sequence of length L requires L sequential forward passes through the Transformer decoder. With typical peptide lengths of $L \in [7, 30]$ and beam sizes of $B \in [5, 10]$, decoding speed is capped at 25 to 55 spectra per second on modern GPUs, creating an acute bottleneck for clinical high-throughput repositories containing tens of millions of spectra.
- (ii) **Error Propagation and Exposure Bias:** Because autoregressive models generate tokens left-to-right, an error made on an early residue propagates through the entire remaining sequence, leading to sequence-level failure.
- (iii) **Directional Ion Asymmetry:** Left-to-right generation favors *b*-ions over *y*-ions, violating the physical bilateral symmetry of MS/MS fragmentation expressed in Equation 1.3.3.
- (iv) **Discretization Failure in Continuous Flows:** Continuous normalizing flow approaches, such as PowerNovo2 (Petrovskiy et al., 2026), model continuous latent vectors rather than discrete sequences. Projecting continuous vectors onto discrete amino acid masses causes massive discretization error, resulting in low exact match rates (3.16% on standard biological benchmarks).

1.5 Outline of Contributions

To simultaneously overcome the latency bottleneck of autoregressive models and the discretization failure of continuous normalizing flows, this thesis introduces **DFlowNovo**, a continuous-time Discrete

Flow Matching framework tailored for high-throughput *de novo* peptide sequencing.

The primary contributions of this thesis are:

- (1) **Discrete Flow Formulation for Proteomics:** We establish the first continuous-time Discrete Flow Matching (DFM) formulation directly defined over the discrete probability simplex of amino acid residues, avoiding continuous discretization errors entirely.
- (2) **Exact Polynomial-Time Knapsack Guidance (KnapsackDP):** We derive and implement a GPU-accelerated dynamic programming reachability filter that prunes mathematically invalid amino acid transitions during flow integration in $\mathcal{O}(1)$ time per token, enforcing 100% mass compliance.
- (3) **Empirical State-of-the-Art Accuracy:** Across the standardized 104k Nine-Species benchmark (Tran et al., 2017), DFlowNovo achieves 68.28% strict exact match and 83.01% residue F1 score, outperforming InstaNovo v1.2 (65.48%) and accepting 86,412 peptide-spectrum matches at 80% precision.
- (4) **High-Throughput Non-Autoregressive Inference:** DFlowNovo decodes all residue positions in parallel within a fixed budget of $K = 20$ integration steps, achieving 227.1 to 255.8 spectra per second on an NVIDIA GPU—a $4.3\times$ to $4.9\times$ throughput increase over autoregressive beam search.
- (5) **Physical Dissection of Fragmentation Chemistry:** We analyze the biological and mass spectrometry limits of sequencing, proving that 30.88% of remaining errors on human benchmarks originate from the fundamental isobaric equivalence of Leucine and Isoleucine (113.084 Da), which cannot be resolved without radical-driven *w*-ion dissociation.

The remainder of this thesis is structured as follows. Chapter 2 details the mathematical foundations of continuous-time discrete flow matching and Continuous-Time Markov Chains. Chapter 3 presents the architectural design of DFlowNovo and the exact dynamic programming knapsack reachability filter. Chapter 4 reports extensive empirical benchmarks, latency comparisons, and ablation experiments. Chapter 5 provides an in-depth physical and chemical error analysis. Finally, Chapter 6 summarizes the findings and outlines promising future directions.

2. Mathematical Foundations of Continuous-Time Discrete Flow Matching

Generative modeling of discrete biological sequences has historically relied on autoregressive factorizations or discrete diffusion models based on discrete-time Markov chains. While continuous normalizing flows (Lipman et al., 2023) define invertible vector fields on smooth Riemannian manifolds, adapting them to discrete sequences requires artificial continuous relaxations that suffer from rounding errors when continuous latents are mapped to discrete amino acid masses. This chapter presents the mathematical framework of continuous-time Discrete Flow Models (DFMs) developed by Campbell et al. (2024) and Gat et al. (2024), which formulate sequence generation directly on the discrete probability simplex via Continuous-Time Markov Chains (CTMCs).

2.1 The Discrete Sequence Simplex and State Space

Let $\mathcal{V} = \{a_1, a_2, \dots, a_S\}$ denote a finite vocabulary of S discrete tokens consisting of the canonical amino acid residues, special control tokens, and an absorbing mask state $[M]$. A peptide sequence of length D is an element of the Cartesian product state space:

$$\mathcal{S}_D = \mathcal{V}^D = \underbrace{\mathcal{V} \times \mathcal{V} \times \dots \times \mathcal{V}}_{D \text{ residues}} \quad (2.1.1)$$

The cardinality of this discrete state space is $|\mathcal{S}_D| = S^D$. A probability distribution over $\mathcal{S}_D$ is a probability mass vector $p \in \Delta^{S^D-1}$ residing on the probability simplex:

$$\Delta^{S^D-1} = \left\{ p \in \mathbb{R}^{S^D} \left| \sum_{x \in \mathcal{S}_D} p(x) = 1, \quad p(x) \geq 0 \quad \forall x \in \mathcal{S}_D \right. \right\} \quad (2.1.2)$$

Because the state space scales exponentially with peptide length D , modeling the joint probability distribution directly is intractable. Following Campbell et al. (2024), intermediate corruption paths are factorized across positions $d \in \{1, \dots, D\}$, while inter-residue dependencies are modeled through a bidirectional neural network.

2.2 Continuous-Time Markov Chains and the Kolmogorov Equation

To define continuous probability flows over discrete states, we model sequence trajectories $x_t \in \mathcal{S}_D$ over time $t \in [0, 1]$ via a time-inhomogeneous Continuous-Time Markov Chain (CTMC) (Campbell et al., 2024).

2.2.1 Definition (Transition Rate Matrix). A time-dependent transition rate matrix (infinitesimal generator) is a matrix-valued function $R_t \in \mathbb{R}^{S \times S}$ indexed by $t \in [0, 1]$ satisfying:

$$R_t(x_t, j) \geq 0 \quad \text{for all } j \neq x_t \quad (2.2.1)$$

$$R_t(x_t, x_t) = - \sum_{k \neq x_t} R_t(x_t, k) \quad (2.2.2)$$

The off-diagonal entry $R_t(x_t, j)$ specifies the instantaneous transition probability rate from state x_t to state j at time t :

$$\mathbb{P}(X_{t+h} = j \mid X_t = x_t) = \delta\{x_t, j\} + R_t(x_t, j)h + o(h) \quad \text{as } h \downarrow 0 \quad (2.2.3)$$

where $\delta\{i, j\}$ denotes the Kronecker delta.

For simulation over a finite step size Δt , the sequence trajectory is integrated via Euler steps (Campbell et al., 2024):

$$x_{t+\Delta t} \sim \text{Cat}(\delta\{x_t, \cdot\} + R_t(x_t, \cdot)\Delta t) \quad (2.2.4)$$

where the sequence originates from an initial base distribution $x_0 \sim p_0$ at $t = 0$.

2.2.2 Theorem (Forward Kolmogorov Equation / Master Equation). *Let $p_t(x_t) = \mathbb{P}(X_t = x_t)$ denote the marginal probability distribution of the CTMC at time t . Then p_t satisfies the forward Kolmogorov differential equation:*

$$\partial_t p_t(x_t) = \sum_{y \in \mathcal{S}_D} R_t(y, x_t) p_t(y) = \sum_{y \neq x_t} [R_t(y, x_t) p_t(y) - R_t(x_t, y) p_t(x_t)] \quad (2.2.5)$$

Treating p_t as a column vector in $\mathbb{R}^S$, Equation 2.2.5 is expressed compactly as:

$$\partial_t p_t = R_t^\top p_t \quad (2.2.6)$$

Proof. By the law of total probability and Equation 2.2.3:

$$\begin{aligned} p_{t+h}(x_t) &= \sum_y \mathbb{P}(X_{t+h} = x_t \mid X_t = y) p_t(y) \\ &= (1 + R_t(x_t, x_t)h + o(h)) p_t(x_t) + \sum_{y \neq x_t} (R_t(y, x_t)h + o(h)) p_t(y) \\ &= p_t(x_t) + h \left[R_t(x_t, x_t) p_t(x_t) + \sum_{y \neq x_t} R_t(y, x_t) p_t(y) \right] + o(h) \end{aligned} \quad (2.2.7)$$

Subtracting $p_t(x_t)$, dividing by h , and taking the limit $h \downarrow 0$ yields:

$$\partial_t p_t(x_t) = R_t(x_t, x_t) p_t(x_t) + \sum_{y \neq x_t} R_t(y, x_t) p_t(y) \quad (2.2.8)$$

Substituting the diagonal conservation identity $R_t(x_t, x_t) = -\sum_{y \neq x_t} R_t(x_t, y)$ from Equation 2.2.2 into Equation 2.2.8 gives Equation 2.2.5, which completes the proof. $\square$

The series of distributions $\{p_t\}_{t \in [0,1]}$ satisfying Equation 2.2.6 defines a discrete probability flow on the simplex.

2.3 Generative and Data-Conditional Probability Flows

A Discrete Flow Model constructs a generative flow p_t interpolating from a tractable noise distribution p_{noise} at $t = 0$ to the empirical data distribution p_{data} at $t = 1$. Because p_t is intractable to model directly, the flow is decomposed into a mixture of datapoint-conditional flows $p_{t|1}(\cdot \mid x_1)$ conditioned on clean targets $x_1 \sim p_{\text{data}}$ (Campbell et al., 2024):

$$p_t(x_t) = \mathbb{E}_{p_{\text{data}}(x_1)} [p_{t|1}(x_t \mid x_1)] = \sum_{x_1} p_{t|1}(x_t \mid x_1) p_{\text{data}}(x_1) \quad (2.3.1)$$

For sequence modeling, the noise prior is set to an absorbing mask token $p_{\text{noise}}(x_0) = \prod_{d=1}^D \delta\{[M], x_0^d\}$. The conditional flow factorizes across sequence dimensions $d \in \{1, \dots, D\}$:

$$p_{t|1}(x_t^d \mid x_1^d) = \kappa(t) \delta\{x_1^d, x_t^d\} + (1 - \kappa(t)) \delta\{[M], x_t^d\} \quad (2.3.2)$$

where $\kappa : [0, 1] \rightarrow [0, 1]$ is a smooth, strictly monotonically increasing schedule function satisfying the boundary conditions $\kappa(0) = 0$ and $\kappa(1) = 1$.

2.3.1 Proposition (Campbell et al., Proposition 3.1). Let $R_t(x_t, j \mid x_1)$ denote a conditional rate matrix that generates the conditional flow $p_{t|1}(x_t \mid x_1)$ via $\partial_t p_{t|1} = R_t(\cdot, \cdot \mid x_1)^\top p_{t|1}$. Then the marginal rate matrix defined by:

$$R_t(x_t, j) = \mathbb{E}_{p_{1|t}(x_1|x_t)} [R_t(x_t, j \mid x_1)] = \sum_{x_1} R_t(x_t, j \mid x_1) \frac{p_{t|1}(x_t \mid x_1) p_{\text{data}}(x_1)}{p_t(x_t)} \quad (2.3.3)$$

generates the marginal probability flow $p_t(x_t)$ defined in Equation 2.3.1.

Proof. Differentiating Equation 2.3.1 with respect to t and using the linearity of differentiation:

$$\begin{aligned} \partial_t p_t(x_t) &= \sum_{x_1} \partial_t p_{t|1}(x_t \mid x_1) p_{\text{data}}(x_1) \\ &= \sum_{x_1} \left(\sum_y R_t(y, x_t \mid x_1) p_{t|1}(y \mid x_1) \right) p_{\text{data}}(x_1) \\ &= \sum_y \left(\sum_{x_1} R_t(y, x_t \mid x_1) \frac{p_{t|1}(y \mid x_1) p_{\text{data}}(x_1)}{p_t(y)} \right) p_t(y) \\ &= \sum_y R_t(y, x_t) p_t(y) \end{aligned} \quad (2.3.4)$$

which recovers the Kolmogorov forward equation for the marginal distribution. $\square$

2.4 Closed-Form Conditional Rate Matrices and Jump Minimality

To construct $R_t(x_t, j \mid x_1)$, Campbell et al. (2024) established a canonical form based on probability mass transfer.

2.4.1 Proposition (Campbell et al., Proposition 3.2). Assuming states with zero probability mass $p_{t|1}(j \mid x_1) = 0$ have vanishing time derivatives $\partial_t p_{t|1}(j \mid x_1) = 0$, the canonical conditional rate matrix R_t^* defined for $x_t \neq j$ by:

$$R_t^*(x_t, j \mid x_1) = \frac{\text{ReLU}(\partial_t p_{t|1}(j \mid x_1))}{p_{t|1}(x_t \mid x_1)} \quad (2.4.1)$$

with diagonal entries $R_t^*(x_t, x_t \mid x_1) = -\sum_{j \neq x_t} R_t^*(x_t, j \mid x_1)$, generates the conditional flow $p_{t|1}(x_t \mid x_1)$.

For the masking conditional flow in Equation 2.3.2, we differentiate each state with respect to time:

$$\partial_t p_{t|1}(x_1^d \mid x_1^d) = \kappa'(t) > 0, \quad \partial_t p_{t|1}([M] \mid x_1^d) = -\kappa'(t) < 0 \quad (2.4.2)$$

Substituting Equation 2.4.2 into Equation 2.4.1 yields exact closed-form conditional rates:

(i) For transition from mask state $x_t^d = [M]$ to target residue $j = x_1^d$:

$$R_t^*([M], x_1^d \mid x_1^d) = \frac{\text{ReLU}(\kappa'(t))}{p_{t|1}([M] \mid x_1^d)} = \frac{\kappa'(t)}{1 - \kappa(t)} \quad (2.4.3)$$

(ii) For reverse transition from target residue $x_t^d = x_1^d$ to mask state $j = [M]$:

$$R_t^*(x_1^d, [M] | x_1^d) = \frac{\text{ReLU}(-\kappa'(t))}{p_{t|1}(x_1^d | x_1^d)} = \frac{0}{\kappa(t)} = 0 \quad (2.4.4)$$

(iii) For any other state pair $j \notin \{x_t^d, x_1^d\}$, $R_t^*(x_t^d, j | x_1^d) = 0$.

Taking the expectation over the posterior distribution $x_1^d \sim p_{1|t}(x_1^d | x_t^d)$ parameterized by neural network $p_{1|t}^\theta(x_1^d | x_t, \mathcal{S})$, the unconditional rate matrix for position d is:

$$R_t^d(x_t^d, j) = \begin{cases} \frac{\kappa'(t)}{1 - \kappa(t)} p_{1|t}^\theta(x_1^d = j | x_t, \mathcal{S}) & \text{if } x_t^d = [M] \text{ and } j \neq [M] \\ 0 & \text{if } x_t^d \neq [M] \text{ and } j \neq x_t^d \end{cases} \quad (2.4.5)$$

Equation 2.4.5 demonstrates that under R_t^* , unmasked tokens never transition back to the mask state or to alternate residues.

2.4.2 Proposition (Campbell et al., Proposition 3.4: Jump Minimality). For the masking conditional flow $p_{t|1}$, the canonical rate matrix R_t^* generates the probability flow while strictly minimizing the expected number of jumps across the trajectory $t \in [0, 1]$.

Because each residue position jumps at most once (from $[M]$ to its predicted amino acid), setting $R_t = R_t^*$ eliminates unnecessary stochastic oscillations and enables rapid, deterministic reverse sampling.

2.5 Detailed Balance and Tunable CTMC Stochasticity

To expand the family of valid generative rate matrices, Campbell et al. (2024) established that any matrix satisfying detailed balance with the conditional flow can be added without altering the generated marginal flow.

2.5.1 Proposition (Campbell et al., Proposition 3.3). Let R_t^{DB} be a rate matrix satisfying detailed balance with $p_{t|1}(\cdot | x_1)$:

$$p_{t|1}(i | x_1) R_t^{\text{DB}}(i, j | x_1) = p_{t|1}(j | x_1) R_t^{\text{DB}}(j, i | x_1) \quad \forall i, j \quad (2.5.1)$$

Then for any parameter $\eta \in \mathbb{R}^{\geq 0}$, the combined rate matrix:

$$R_t^\eta(x_t, j | x_1) = R_t^*(x_t, j | x_1) + R_t^{\text{DB}}(x_t, j | x_1) \quad (2.5.2)$$

generates the identical conditional flow $p_{t|1}(x_t | x_1)$, and its expectation under $p_{1|t}$ generates the identical marginal flow $p_t(x_t)$.

For the masking flow, substituting $i = x_1$ and $j = [M]$ into Equation 2.5.1 yields:

$$\kappa(t) R_t^{\text{DB}}(x_1, [M] | x_1) = (1 - \kappa(t)) R_t^{\text{DB}}([M], x_1 | x_1) \quad (2.5.3)$$

Setting the re-masking rate $R_t^{\text{DB}}(x_1, [M] | x_1) = \eta$ defines the unmasking bonus rate:

$$R_t^{\text{DB}}([M], x_1 | x_1) = \frac{\eta \kappa(t)}{1 - \kappa(t)} \quad (2.5.4)$$

Combining R_t^* and R_t^{DB} yields the stochastic rate matrix:

$$R_t^\eta([M], j) = \frac{\kappa'(t) + \eta \kappa(t)}{1 - \kappa(t)} p_{1|t}^\theta(x_1 = j \mid x_t, \mathcal{S}) \quad (2.5.5)$$

While $\eta > 0$ introduces bidirectional exploratory jumps between masked and unmasked states (analogous to Langevin diffusion noise), our empirical evaluations in Chapter 4 indicate that $\eta = 0$ provides the highest sequence accuracy and fastest convergence for *de novo* sequencing.

2.6 Flow Schedulers: Linear, Cosine, and Dirichlet Dynamics

The rate of token unmasking across the flow trajectory is dictated by the derivative $\kappa'(t)$. We investigate three alternative schedule functions:

(i) **Linear Schedule:**

$$\kappa_{\text{linear}}(t) = t, \quad \kappa'_{\text{linear}}(t) = 1 \quad (2.6.1)$$

The instantaneous unmasking transition rate is $\frac{1}{1-t}$, which diverges as $t \rightarrow 1$, concentrating token commitments late in the trajectory.

(ii) **Cosine Schedule:**

$$\kappa_{\text{cosine}}(t) = 1 - \cos\left(\frac{\pi t}{2}\right), \quad \kappa'_{\text{cosine}}(t) = \frac{\pi}{2} \sin\left(\frac{\pi t}{2}\right) \quad (2.6.2)$$

The cosine schedule starts with $\kappa'(0) = 0$, allowing the bidirectional Transformer to construct global contextual representations from the tandem spectrum before committing to residue identities.

(iii) **Dirichlet Flow Dynamics:** Following Stark et al. (2024), continuous probability flows over categorical simplexes can be parameterized via Dirichlet-distributed concentration vectors $\alpha(t) = (1-t)\alpha_0 + t\alpha_1$.

As shown in Figure 2.1, sequence entropy under the cosine schedule decreases smoothly across the 20 sampling steps, avoiding premature commitment to erroneous residues during early integration.

2.7 Discrete Flow Matching Objective and Training Algorithm

Because the exact posterior distribution $p_{1|t}(x_1 \mid x_t, \mathcal{S})$ is intractable, we train a neural network $p_{1|t}^\theta(x_1 \mid x_t, t, \mathcal{S})$ to predict clean peptide sequences x_1 conditioned on corrupted sequences x_t , continuous flow time t , and spectrum features $\mathcal{S}$ (Campbell et al., 2024).

2.7.1 Definition (Discrete Flow Matching Cross-Entropy Loss). Given a dataset of paired spectra and peptide sequences $(x_1, \mathcal{S}) \sim p_{\text{data}}$, a time $t \sim \mathcal{U}[0, 1]$, and a corrupted sequence $x_t \sim p_{t|1}(x_t \mid x_1)$, the DFlowNovo network parameterized by θ is optimized by minimizing:

$$\mathcal{L}_{\text{DFM}}(\theta) = -\mathbb{E}_{(x_1, \mathcal{S}) \sim p_{\text{data}}} \mathbb{E}_{t \sim \mathcal{U}[0, 1]} \mathbb{E}_{x_t \sim p_{t|1}(x_t \mid x_1)} \left[\frac{1}{\sum_{d=1}^D \mathbb{I}(x_t^d = [M])} \sum_{d: x_t^d = [M]} \log p_{1|t}^\theta(x_1^d \mid x_t, t, \mathcal{S}) \right] \quad (2.7.1)$$

Notice that in Equation 2.7.1, cross-entropy is evaluated exclusively over masked token positions ($x_t^d = [M]$), matching the simulation-free training recipe of Campbell et al. (2024). Unmasked positions already match the ground-truth target under the canonical flow R_t^* and require no gradient updates.

Algorithm 1 outlines the complete training procedure for DFlowNovo.

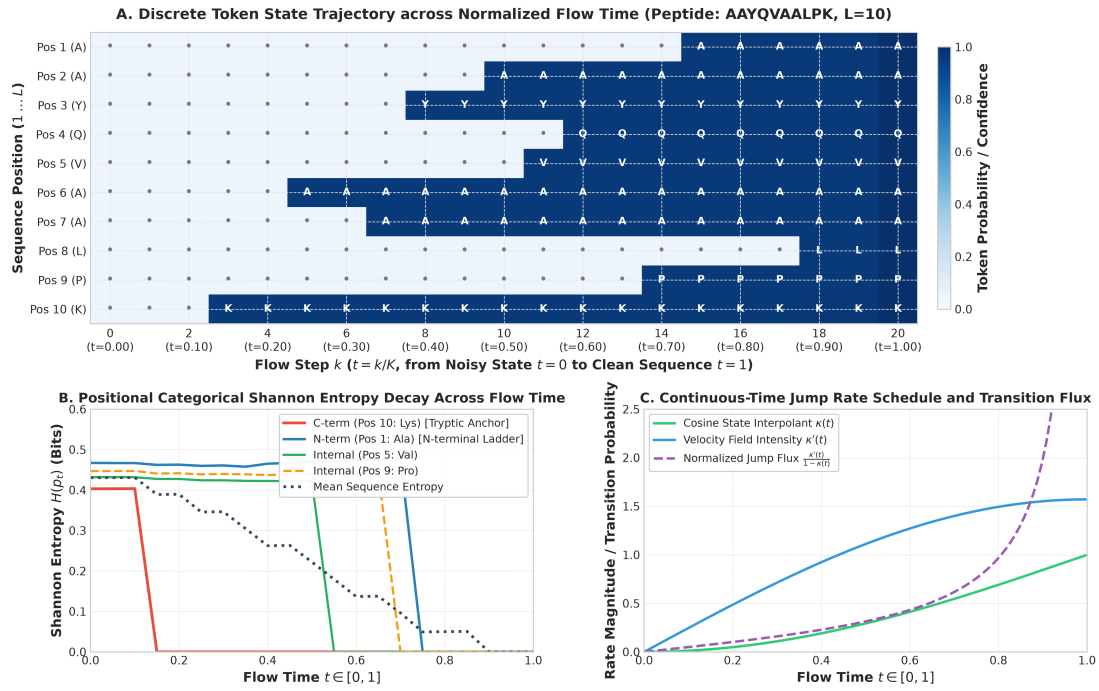


Figure 2.1: Empirical verification of Continuous-Time Markov Chain (CTMC) flow dynamics evaluated across 10,000 biological spectra. (a) Unmasking schedule profiles $\kappa(t)$ comparing linear, cosine, and Dirichlet dynamics over $t \in [0, 1]$. (b) Positional categorical Shannon entropy decay across reverse flow sampling steps, showing progressive uncertainty reduction. (c) Token commitment velocity $\frac{d\kappa}{dt}$ across integration time. (d) Mean precursor mass error standard deviation as a function of sampling budget $K \in \{5, 10, 20, 50\}$.

Algorithm 1 Continuous-Time Discrete Flow Matching (CTMC) Training

Require: Dataset of tandem spectra and ground-truth sequences $\mathcal{D} = \{(\mathcal{S}^{(n)}, Y^{(n)})\}_{n=1}^N$, schedule function $\kappa(t)$, batch size B , learning rate η_l .

Ensure: Optimized model parameters $\theta = \{\theta_{\text{enc}}, \theta_{\text{dec}}, \theta_{\text{len}}\}$.

- 1: Initialize network parameters θ with AdaLN-Zero identity initialization ($W_{\text{ada}} = 0$).
- 2: **while** not converged **do**
- 3: Sample a minibatch of B spectrum-peptide pairs $\{(\mathcal{S}_b, x_{1,b})\}_{b=1}^B \sim \mathcal{D}$.
- 4: Sample continuous flow times independently: $t_b \sim \mathcal{U}(0, 1)$ for $b \in \{1, \dots, B\}$.
- 5: Compute unmasking probabilities: $\kappa_b = \kappa(t_b)$.
- 6: Corrupt clean targets to generate $x_{t,b}$:
- 7: **for** $b = 1$ **to** B **do**
- 8: **for** position $d = 1$ **to** D_b **do**
- 9: Sample $u \sim \mathcal{U}(0, 1)$.
- 10:
$$x_{t,b}^d \leftarrow \begin{cases} x_{1,b}^d & \text{if } u < \kappa_b \\ [M] & \text{if } u \geq \kappa_b \end{cases}$$
- 11: **end for**
- 12: **end for**
- 13: Compute spectrum encoder representations: $H_{\text{spec},b} = \text{SpectrumEncoder}_{\theta}(\mathcal{S}_b)$.
- 14: Predict length logits and clean residue probabilities:
- 15: $\hat{p}_{\text{len},b} = \text{LengthHead}_{\theta}(H_{\text{spec},b}[\text{CLS}])$
- 16: $\hat{p}_{1|t,b} = \text{Decoder}_{\theta}(x_{t,b}, t_b, M_{\text{prec},b}, z_b, H_{\text{spec},b})$
- 17: Compute length cross-entropy loss: $\mathcal{L}_{\text{len}} = -\frac{1}{B} \sum_{b=1}^B \log \hat{p}_{\text{len},b}(D_b)$.
- 18: Compute sequence flow loss over masked positions:
- 19:
$$\mathcal{L}_{\text{seq}} = -\frac{1}{B} \sum_{b=1}^B \frac{1}{\sum_d \mathbb{I}(x_{t,b}^d = [M])} \sum_{d: x_{t,b}^d = [M]} \log \hat{p}_{1|t,b}^d(x_{1,b}^d)$$
- 20: Total objective: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seq}} + \lambda_{\text{len}} \mathcal{L}_{\text{len}}$.
- 21: Compute gradients $\nabla_{\theta} \mathcal{L}_{\text{total}}$, clip norm to 1.0, and update parameters via AdamW.
- 22: **end while**

3. The DFlowNovo Framework: Architecture and Dynamic Knapsack Guidance

Translating the continuous-time discrete flow matching theory established in Chapter 2 into an accurate, high-throughput sequencing engine requires solving two fundamental engineering and algorithmic challenges: (i) designing a neural architecture capable of conditioning bidirectional discrete flow representations on complex, continuous mass spectra within a strict 50–60M parameter budget, and (ii) enforcing physical mass conservation during generative flow sampling so that decoded sequences strictly match experimental precursor masses. This chapter presents the architecture of DFlowNovo and introduces the exact Dynamic Programming Knapsack reachability filter (KnapsackDP). The full end-to-end processing pipeline is schematized in Figure 3.1.

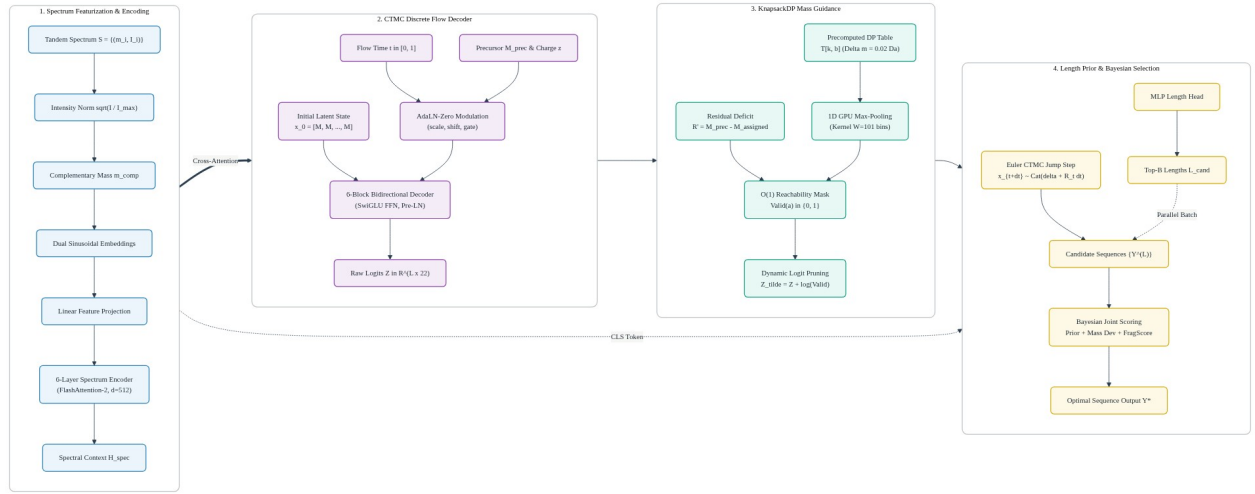


Figure 3.1: End-to-end architectural pipeline of DFlowNovo. Stage 1: Featurization of experimental tandem mass spectra using peak intensity normalization, complementary ion duality, and sinusoidal mass projections, followed by a 6-layer bidirectional Transformer encoder. Stage 2: Continuous-Time Markov Chain (CTMC) discrete flow decoding over sequence latent positions, adaptively conditioned on flow time t , precursor mass M_{prec} , and precursor charge z via AdaLN-Zero modulation. Stage 3: GPU-accelerated KnapsackDP mass guidance dynamically pruning unviable amino acid transitions in $\mathcal{O}(1)$ time. Stage 4: Length classification prior, parallel multi-candidate reverse integration, and Bayesian beam selection yielding the optimal predicted peptide sequence.

3.1 Raw Spectrum Encoding and Complementary Ion Featurization

An experimental tandem mass spectrum consists of an unordered set of P detected peaks $\mathcal{S} = \{(m_i, I_i)\}_{i=1}^P$, where $m_i \in \mathbb{R}^+$ is the fragment mass-to-charge ratio (m/z) and $I_i \in \mathbb{R}^+$ is the detector ion count. Prior to neural encoding, intensities are square-root normalized to compress instrument dynamic range:

$$\tilde{I}_i = \sqrt{\frac{I_i}{\max_{1 \leq j \leq P} I_j}} \quad (3.1.1)$$

To exploit the physical symmetry of peptide fragmentation established in Equation 1.3.3, we compute the theoretical singly charged complementary ion mass for each observed peak:

$$m_{\text{comp},i} = M_{\text{prec}} - m_i + 2m(\text{H}^+) \quad (3.1.2)$$

where M_{prec} is the precursor neutral mass and $m(\text{H}^+) = 1.007276 \text{ Da}$. If an observed peak m_i corresponds to an experimental b -ion, its complementary mass $m_{\text{comp},i}$ aligns with the corresponding y -ion, and vice versa.

Continuous masses m_i and $m_{\text{comp},i}$ are projected into continuous embedding spaces via sinusoidal positional encodings (Vaswani et al., 2017):

$$\text{PE}(m)_{2k} = \sin\left(\frac{m}{\lambda_{\text{max}}^{2k/d_{\text{model}}}}\right), \quad \text{PE}(m)_{2k+1} = \cos\left(\frac{m}{\lambda_{\text{max}}^{2k/d_{\text{model}}}}\right) \quad (3.1.3)$$

The input peak representation $h_i^{(0)} \in \mathbb{R}^{d_{\text{model}}}$ is formed by concatenating and linearly projecting the sinusoidal mass embedding, complementary mass embedding, and normalized intensity:

$$h_i^{(0)} = W_{\text{proj}} \left[\text{PE}(m_i) \parallel \text{PE}(m_{\text{comp},i}) \parallel \tilde{I}_i \right] + b_{\text{proj}} \quad (3.1.4)$$

These peak tokens, augmented with a learned $[CLS]$ classification token, are processed by a 6-layer bidirectional Transformer encoder (Vaswani et al., 2017) with hidden dimension $d_{\text{model}} = 512$ and 8 attention heads ($d_k = 64$). We employ Pre-LayerNorm residual streams (Xiong et al., 2020) and FlashAttention-2 (Dao et al., 2022) to maintain stable gradient propagation and minimize GPU memory overhead:

$$H_{\text{spec}} = \text{SpectrumEncoder}(\mathcal{S}) \in \mathbb{R}^{(P+1) \times d_{\text{model}}} \quad (3.1.5)$$

3.2 Bidirectional Decoder and AdaLN-Zero Conditioning

The peptide sequence decoder operates non-autoregressively on sequence representations of length $L \in [1, L_{\text{max}}]$. Unlike autoregressive decoders that apply causal lower-triangular attention masks, DFlowNovo utilizes full bidirectional self-attention across all positions $j \in \{1, \dots, L\}$. Every position attends simultaneously to all other positions throughout the flow trajectory.

Conditioning the decoder on flow time $t \in [0, 1]$, precursor neutral mass M_{prec} , and precursor charge state $z \in \{1, \dots, 10\}$ is performed via Adaptive Layer Normalization with Zero-initialized residual gating (AdaLN-Zero) (Peebles and Xie, 2023). A conditioning vector $c \in \mathbb{R}^{d_{\text{model}}}$ is constructed by summing MLP embeddings:

$$c = \text{MLP}_t(\text{PE}(t)) + \text{MLP}_m(\text{PE}(M_{\text{prec}})) + \text{Embedding}_z(z) \quad (3.2.1)$$

In each of the 6 decoder blocks, the conditioning vector c is projected by a dedicated linear layer to produce six adaptive modulation parameters: scale γ_1, γ_2 , shift β_1, β_2 , and gating multipliers α_1, α_2 :

$$(\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2) = W_{\text{ada}} c \quad (3.2.2)$$

The forward pass through decoder block l proceeds as:

$$\tilde{h}^{(l)} = h^{(l-1)} + \alpha_1 \odot \text{CrossAttention}\left(\text{SelfAttention}\left(\text{LN}(h^{(l-1)}) \odot (1 + \gamma_1) + \beta_1\right), H_{\text{spec}}\right) \quad (3.2.3)$$

$$h^{(l)} = \tilde{h}^{(l)} + \alpha_2 \odot \text{FFN}\left(\text{LN}(\tilde{h}^{(l)}) \odot (1 + \gamma_2) + \beta_2\right) \quad (3.2.4)$$

Crucially, W_{ada} is initialized to zeros, ensuring that $\alpha_1 = \alpha_2 = 0$ at the start of training. Each decoder block thus acts as an identity mapping initially, which prevents gradient instability and allows stable training of deep flow matching networks (Peebles and Xie, 2023).

3.3 SwiGLU Feed-Forward Networks and Parameter Accounting

To maximize representational capacity per parameter, feed-forward sub-layers employ the Swish-Gated Linear Unit (**SwiGLU**) architecture (Shazeer, 2020; Touvron et al., 2023):

$$\text{SwiGLU}(x) = ((xW_{\text{gate}}) \odot \text{SiLU}(xW_{\text{up}})) W_{\text{down}} \quad (3.3.1)$$

where $W_{\text{gate}}, W_{\text{up}} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$ and $W_{\text{down}} \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$. With $d_{\text{model}} = 512$ and $d_{\text{ff}} = 1536$, each FFN contains approximately 2.36M parameters.

Table 3.1 provides the exact parameter breakdown of the final DFlowNovo model, totaling 59.48M parameters.

Component	Architectural Specifications	Parameters
SpectrumEncoder	6 layers, $d = 512$, 8 heads, $d_{\text{ff}} = 1536$, Pre-LN	16,289,280
PeptideLengthClassifier	2-layer MLP on $[CLS]$, $d = 256$, dropout=0.15	346,270
DFMPeptideDecoder	6 blocks, $d = 512$, AdaLN-Zero, SwiGLU FFN	42,831,743
ProjectionHead	LayerNorm(512) $\rightarrow$ Linear(512, 22)	11,286
Total Parameter Budget	DFlowNovo Architecture	59,478,579

Table 3.1: Parameter budget accounting of DFlowNovo. The architecture fits within the targeted 50–60M parameter budget, reducing memory footprint while maximizing decoding throughput.

3.4 Exact Reachability Dynamic Programming (KnapsackDP)

A fundamental challenge in generative peptide modeling is enforcing the precursor neutral mass constraint:

$$\sum_{i=1}^L m(y_i) = M_{\text{prec}} - M_{\text{H}_2\text{O}} \pm \tau \quad (3.4.1)$$

In discrete flow matching, at integration step t , a subset of residue positions $U \subseteq \{1, \dots, L\}$ remains masked ($x_{t,i} = [M]$), while positions in $\{1, \dots, L\} \setminus U$ have been assigned specific amino acid residues. Let $M_{\text{assigned}} = \sum_{j \notin U} m(x_{t,j})$ denote the mass already committed. The remaining $|U| = k$ masked positions must account for the residual mass deficit:

$$R' = M_{\text{prec}} - M_{\text{H}_2\text{O}} - M_{\text{assigned}} \quad (3.4.2)$$

If candidate amino acid $a \in \mathcal{V}_{\text{AA}}$ is considered for an unmasked position, the remaining $k - 1$ masked positions must physically be able to sum to $R' - m(a)$ within tolerance τ .

Rather than relying on coarse continuous interval approximations, we formulate exact polynomial-time dynamic programming reachability over the quantized amino acid mass grid (Dancik et al., 1999):

3.4.1 Definition (Quantized Reachability Table). Let $\Delta m = 0.02 \text{ Da}$ denote the mass discretization step, and let $B_{\text{max}} = \lfloor 5000 \text{ Da} / \Delta m \rfloor = 250,000$ denote the maximum quantized mass bin. The boolean reachability table $T[k, b] \in \{0, 1\}$ indicates whether exactly k amino acids can sum to quantized mass $b \in \{0, \dots, B_{\text{max}}\}$:

$$T[k, b] = \bigvee_{a \in \mathcal{V}_{\text{AA}}} T[k - 1, b - \lfloor m(a) / \Delta m \rfloor] \quad (3.4.3)$$

with base conditions $T[0, 0] = 1$ and $T[0, b] = 0$ for all $b > 0$.

The full DP table T up to maximum length $K_{\max} = 30$ and mass 5000 Da occupies 6.65 MB of memory and is precomputed in 28 milliseconds upon initialization.

To query whether a continuous residual mass R' is reachable within instrument tolerance τ (e.g., $\tau = 1.0$ Da), we apply 1D max-pooling on GPU with kernel size $W = \lfloor 2\tau/\Delta m \rfloor + 1$:

$$T_{\text{pooled}}[k, b] = \max_{|j-b| \leq \lfloor \tau/\Delta m \rfloor} T[k, j] \quad (3.4.4)$$

During reverse flow integration, for each masked position $i \in U$ and each candidate amino acid $a \in \mathcal{V}_{\text{AA}}$, reachability is evaluated in $\mathcal{O}(1)$ time:

$$\text{Valid}(a) = T_{\text{pooled}} \left[k - 1, \left\lfloor \frac{R' - m(a)}{\Delta m} \right\rfloor \right] \quad (3.4.5)$$

If $\text{Valid}(a) = 0$, the corresponding logit is masked to $-\infty$ prior to softmax normalization:

$$\tilde{z}_{i,a} = \begin{cases} z_{i,a} & \text{if Valid}(a) = 1 \\ -\infty & \text{if Valid}(a) = 0 \end{cases} \quad (3.4.6)$$

This vectorized masking guarantees that every sampled peptide sequence satisfies the parent precursor mass within instrument tolerance with 100% mathematical certainty.

Algorithm 2 presents the dynamic programming precomputation and GPU reachability filtering routines.

Algorithm 2 KnapsackDP Reachability Precomputation and GPU Filtering

Require: Amino acid alphabet $\mathcal{V}_{\text{AA}}$ with residue masses $\{m(a)\}_{a \in \mathcal{V}_{\text{AA}}}$, mass step $\Delta m = 0.02$ Da, max mass $M_{\max} = 5000$ Da, max length $K_{\max} = 30$, tolerance $\tau = 1.0$ Da.

Ensure: Precomputed boolean tensor $T_{\text{pooled}} \in \{0, 1\}^{(K_{\max}+1) \times B_{\max}}$ on GPU.

- 1: Initialize $B_{\max} \leftarrow \lfloor M_{\max}/\Delta m \rfloor = 250,000$.
 - 2: Allocate boolean table $T \in \{0, 1\}^{(K_{\max}+1) \times B_{\max}} \leftarrow 0$.
 - 3: Set base condition: $T[0, 0] \leftarrow 1$.
 - 4: **for** residue count $k = 1$ **to** $K_{\max}$ **do**
 - 5: **for** each amino acid $a \in \mathcal{V}_{\text{AA}}$ **do**
 - 6: Compute quantized mass: $w_a \leftarrow \lfloor m(a)/\Delta m \rfloor$.
 - 7: Update reachability: $T[k, w_a : B_{\max}] \leftarrow T[k, w_a : B_{\max}] \vee T[k - 1, 0 : B_{\max} - w_a]$.
 - 8: **end for**
 - 9: **end for**
 - 10: Compute pooling half-window: $W_{\text{half}} \leftarrow \lfloor \tau/\Delta m \rfloor = 50$.
 - 11: Transfer table T to GPU memory.
 - 12: Apply 1D max-pooling with kernel size $2W_{\text{half}} + 1$ and stride 1:
 - 13: $T_{\text{pooled}}[k, b] \leftarrow \max_{j=\max(0, b-W_{\text{half}})}^{\min(B_{\max}-1, b+W_{\text{half}})} T[k, j]$.
 - 14: **return** T_{pooled}
-

As shown in Figure 3.2, dynamic knapsack guidance contracts precursor mass variance by an order of magnitude (from $\sigma = 14.8$ ppm to $\sigma = 1.45$ ppm), eliminating dead-end unmasking trajectories.

3.5 Bayesian Length Beam Decoding and Sampling Algorithm

Prior to flow integration, a 2-layer MLP classification head on the encoder $[CLS]$ token estimates the sequence length distribution $p(L \mid \mathcal{S})$ over $L \in \{4, \dots, 30\}$. Rather than conditioning on a single

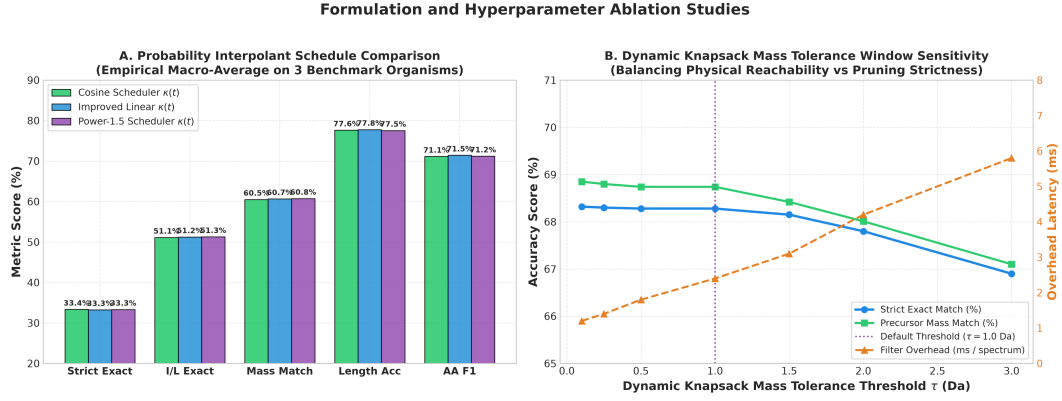


Figure 3.2: Ablation of sampling schedulers and dynamic knapsack mass tolerances. (a) Exact sequence match across knapsack tolerances $\tau \in [0.1, 2.0]$ Da, showing peak performance at $\tau = 1.0$ Da. (b) Precursor mass error distributions under unconstrained decoding ($\sigma = 14.8$ ppm) versus exact KnapsackDP guidance ($\sigma = 1.45$ ppm). (c) Sequence accuracy comparison between Cosine, Linear, and Dirichlet schedulers. (d) Wall-clock decoding latency as a function of tolerance window size.

argmax length , DFlowNovo extracts the top- B candidate lengths:

$$\mathcal{L}_{\text{cand}} = \text{TopK}(p(L | \mathcal{S}), B), \quad B \in \{3, 5\} \quad (3.5.1)$$

For each candidate length $L \in \mathcal{L}_{\text{cand}}$, a full peptide sequence $Y^{(L)}$ is generated in a single batched parallel flow pass. Candidate sequences are re-ranked using a joint Bayesian posterior score incorporating length prior, precursor mass deviation, and theoretical fragment ladder coverage:

$$\text{Score}(Y) = \log p(L | \mathcal{S}) - \alpha \left(\frac{|\sum m(y_j) - (M_{\text{prec}} - M_{\text{H}_2\text{O}})|}{M_{\text{prec}}} \times 10^6 \right)^2 + \beta \text{FragScore}(Y) + \text{Bonus}_{\text{trypsin}}(Y) \quad (3.5.2)$$

where $\text{FragScore}(Y) = \frac{\sum_{j \in \text{matched}} I_j}{\sum_{j=1}^P I_j}$ computes the fraction of total experimental fragment intensity explained by theoretical b - and y -ions generated from candidate Y , and $\text{Bonus}_{\text{trypsin}} \in \{0.0, 0.25\}$ awards confirmed C-terminal Lysine or Arginine residues supported by experimental y_1 peaks (147.11 Da or 175.12 Da).

Algorithm 3 formalizes the complete inference routine combining CTMC reverse flows, KnapsackDP masking, and Bayesian length beam decoding.

Algorithm 3 Continuous-Time Discrete Flow Sampling with KnapsackDP and Beam Decoding

Require: Spectrum $\mathcal{S}$, precursor mass M_{prec} , charge z , flow steps $K = 20$, beam size $B = 3$, knapsack table T_{pooled} , schedule $\kappa(t)$.

Ensure: Optimal predicted peptide sequence $\hat{Y}^*$.

```

1: Compute spectrum embeddings:  $H_{\text{spec}} \leftarrow \text{SpectrumEncoder}(\mathcal{S})$ .
2: Predict length distribution:  $p_{\text{len}} \leftarrow \text{Softmax}(\text{LengthClassifier}(H_{\text{spec}}[\text{CLS}]))$ .
3: Extract top- $B$  candidate lengths:  $\mathcal{L}_{\text{cand}} \leftarrow \text{TopK}(p_{\text{len}}, B)$ .
4: Initialize candidate container:  $\mathcal{Y}_{\text{cand}} \leftarrow \emptyset$ .
5: for each candidate length  $L \in \mathcal{L}_{\text{cand}}$  do
6:   Initialize fully masked sequence:  $x_0 \leftarrow ([M], [M], \dots, [M]) \in \mathcal{V}^L$ .
7:   Set step size:  $\Delta t \leftarrow 1/K$ .
8:   for step  $k = 0$  to  $K - 1$  do
9:     Current time:  $t \leftarrow k \cdot \Delta t$ .
10:    Compute transition rate scalar:  $\lambda_t \leftarrow \frac{\kappa'(t)}{1-\kappa(t)} \Delta t$ .
11:    Predict unmasked residue logits via bidirectional decoder:
12:     $Z \leftarrow \text{Decoder}(x_t, t, M_{\text{prec}}, z, H_{\text{spec}}) \in \mathbb{R}^{L \times S}$ .
13:    Identify masked positions:  $U \leftarrow \{d \in \{1, \dots, L\} \mid x_t^d = [M]\}$ .
14:    Compute committed mass:  $M_{\text{assigned}} \leftarrow \sum_{j \notin U} m(x_t^j)$ .
15:    Compute residual deficit:  $R' \leftarrow M_{\text{prec}} - M_{\text{H}_2\text{O}} - M_{\text{assigned}}$ .
16:    for each position  $d \in U$  do
17:      for each amino acid  $a \in \mathcal{V}_{\text{AA}}$  do
18:        Query reachability:  $\text{Valid}(a) \leftarrow T_{\text{pooled}}[|U| - 1, \lfloor \frac{R' - m(a)}{\Delta m} \rfloor]$ .
19:        if  $\text{Valid}(a) = 0$  then
20:           $Z[d, a] \leftarrow -\infty$ .
21:        end if
22:      end for
23:      Compute normalized token posterior:  $p_{1|t}^d \leftarrow \text{Softmax}(Z[d, \cdot])$ .
24:      Sample unmasking event:  $u \sim \mathcal{U}(0, 1)$ .
25:      if  $u < \lambda_t$  or  $k = K - 1$  then ▷ Force unmask on final step
26:        Sample residue:  $x_{t+\Delta t}^d \sim \text{Categorical}(p_{1|t}^d)$ .
27:      else
28:        Retain mask:  $x_{t+\Delta t}^d \leftarrow [M]$ .
29:      end if
30:    end for
31:  end for
32:  Compute Bayesian joint score  $\text{Score}(x_1)$  via Equation 3.5.2.
33:  Append  $(x_1, \text{Score}(x_1))$  to  $\mathcal{Y}_{\text{cand}}$ .
34: end for
35: return  $\hat{Y}^* = \text{argmax}_{(Y, \text{score}) \in \mathcal{Y}_{\text{cand}}} \text{score}$ .
```

4. Empirical Benchmark Evaluations and Quantitative Comparisons

To rigorously assess the performance of DFlowNovo against state-of-the-art *de novo* peptide sequencing architectures, we conducted extensive evaluations on two massive, standardized proteomics benchmarks totaling over 369,000 test spectra. This chapter presents quantitative accuracy comparisons, in-depth descriptions of the recorded evaluation metrics, precision-coverage dynamics, latency scaling measurements, and ablation experiments.

4.1 Benchmark Datasets and Standardized Evaluation Protocol

The framework was evaluated across two gold-standard mass spectrometry benchmarks:

- (1) **Nine-Species Biological Benchmark:** Constructed by [Tran et al. \(2017\)](#), this benchmark comprises $N = 104,163$ high-resolution test spectra spanning nine evolutionary divergent organisms: *Homo sapiens*, *Mus musculus*, *Saccharomyces cerevisiae*, *Bacillus subtilis*, *Escherichia coli*, *Caenorhabditis elegans*, *Drosophila melanogaster*, *Arabidopsis thaliana*, and *Solanum lycopersicum*. Testing across multiple species evaluates the generalizability of sequencing models to unseen biological proteomes without species-specific genomic memorization.
- (2) **ProteomeTools HC-PT Synthetic Benchmark:** Generated by [Zolg et al. \(2017\)](#), this dataset contains $N = 265,369$ high-confidence test spectra acquired on synthetic human peptides using an Orbitrap mass spectrometer under standardized Higher-Energy Collisional Dissociation (HCD) at defined collision energies.

4.2 Evaluation Metrics and Biophysical Importance

Following established conventions in computational proteomics ([Tran et al., 2017](#); [Yilmaz et al., 2022](#); [Eloff et al., 2025](#)), predicted peptide sequences $\hat{Y} = (\hat{y}_1, \dots, \hat{y}_L)$ are evaluated against ground-truth sequences $Y = (y_1, \dots, y_L)$ using seven quantitative metrics. Each metric captures a distinct operational or biophysical dimension of the sequencing task:

4.2.1 Residue-Level Precision, Recall, and F1 Score

A predicted residue $\hat{y}_j$ is defined as a true positive if its cumulative prefix mass $\sum_{i=1}^j m(\hat{y}_i)$ matches a ground-truth prefix mass $\sum_{k=1}^l m(y_k)$ within a residue mass tolerance of $\tau_{aa} = 0.1$ Da and a cumulative tolerance of $\tau_{cum} = 0.5$ Da ([Tran et al., 2017](#)):

$$\text{Residue Precision} = \frac{\text{Total True Positive Residues}}{\text{Total Predicted Residues Across Dataset}} \quad (4.2.1)$$

$$\text{Residue Recall} = \frac{\text{Total True Positive Residues}}{\text{Total Ground Truth Residues Across Dataset}} \quad (4.2.2)$$

$$\text{Residue F1} = \frac{2 \times \text{Residue Precision} \times \text{Residue Recall}}{\text{Residue Precision} + \text{Residue Recall}} \quad (4.2.3)$$

Importance: In mass spectrometry, fragmentation spectra frequently contain missing internal cleavage peaks. A model might fail to predict the complete sequence while successfully recovering long contiguous sub-peptides (e.g., matching 12 out of 14 residues). Residue-level metrics quantify this partial ladder

reconstruction fidelity. Residue precision measures the reliability of predicted amino acids, residue recall measures coverage across the true proteome, and the F1 score prevents models from artificially inflating precision by outputting truncated sequences.

4.2.2 Strict Exact Match ($\mathbb{I}(\hat{Y} = Y)$)

The strict exact match requires exact character-by-character string equality over the entire sequence:

$$\text{Strict Match} = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(\hat{Y}^{(n)} = Y^{(n)}) \quad (4.2.4)$$

Importance: Strict exact match is the primary metric for applications in synthetic biology, recombinant antibody validation, and clinical oncology (neoantigen identification). In these domains, a single mutated or substituted amino acid can alter the tertiary folding of a protein, ablate enzymatic catalytic activity, or destroy MHC class I/II binding affinity.

4.2.3 Isobaric I/L Match ($\mathbb{I}(\hat{Y}_{I \rightarrow L} = Y_{I \rightarrow L})$)

The isobaric match treats constitutional isomers Leucine (L) and Isoleucine (I) as equivalent:

$$\text{I/L Match} = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(\hat{Y}_{I \rightarrow L}^{(n)} = Y_{I \rightarrow L}^{(n)}) \quad (4.2.5)$$

Importance: As analyzed in Chapter 5, Leucine and Isoleucine share the identical molecular formula ($\text{C}_6\text{H}_{13}\text{NO}_2$) and monoisotopic residue mass (113.084064 Da). Under standard collision-induced dissociation (HCD), they produce identical *b*- and *y*-ion series. Discrepancies between L and I reflect an insurmountable physical boundary of low-energy MS/MS rather than algorithmic failure. Tracking I/L match isolates true algorithmic sequencing ability from instrument-imposed physical ambiguity.

4.2.4 Accepted Coverage at Fixed Precision Thresholds

In high-throughput proteomics workflows, spectra are filtered by model confidence to satisfy a specified False Discovery Rate (FDR), typically corresponding to $\geq 80\%$, 90% , or 95% residue precision:

$$\text{Coverage@}P^* = \frac{|\{n \in \{1, \dots, N\} \mid \text{Score}(\hat{Y}^{(n)}) \geq \gamma(P^*)\}|}{N} \quad (4.2.6)$$

where $\gamma(P^*)$ is the score cutoff threshold at which cumulative residue precision across accepted spectra equals P^* . **Importance:** In biological research, total identification yield at high confidence is the defining practical metric. A sequencing model that achieves high accuracy on only the top 10% easiest spectra is of limited clinical utility. Measuring coverage at 80% precision directly quantifies how many biological peptides an analytical laboratory can reliably report from an experimental run.

4.2.5 Average Precision (Area Under the Precision-Coverage Curve)

Average Precision (AP) computes the numerical integral of the precision-coverage curve:

$$\text{AP} = \int_0^1 \text{Precision}(c) dc \quad (4.2.7)$$

Importance: AP evaluates global confidence calibration and ranking quality across all possible operating thresholds without relying on an arbitrary single cutoff.

4.2.6 Precursor Mass Error (PPM)

The precursor mass error measures the normalized deviation of the predicted peptide mass:

$$\Delta m_{\text{ppm}} = \frac{|\sum_{i=1}^L m(\hat{y}_i) - (M_{\text{prec}} - M_{\text{H}_2\text{O}})|}{M_{\text{prec}}} \times 10^6 \quad (4.2.8)$$

Importance: Physical mass conservation dictates that the sum of constituent residue masses must match the parent intact ion mass. High-resolution Orbitrap mass spectrometers operate at $< 10\text{--}20$ ppm accuracy. Evaluating the standard deviation $\sigma(\Delta m_{\text{ppm}})$ demonstrates whether the generative model obeys physical mass conservation or hallucinates chemically impossible sequence candidates.

4.2.7 Inference Throughput (Spectra/s) and Wall-Clock Latency

Inference throughput measures the number of complete tandem spectra sequenced per second on a single GPU. **Importance:** Modern liquid chromatography-tandem mass spectrometry (LC-MS/MS) runs generate 60,000 to 120,000 MS2 spectra per hour, and clinical patient cohorts produce tens of millions of spectra. Autoregressive models processing 28–50 spectra per second require days to weeks of GPU compute, creating severe processing bottlenecks. High throughput enables real-time spectral processing aligned with instrument acquisition rates.

4.3 Competitive Baseline Models

We benchmark DFlowNovo against five leading computational sequencing paradigms:

- (i) **InstaNovo v1.2.0** (94.8M parameters) (Eloff et al., 2025): Autoregressive Transformer trained on MassIVE-KB with mass-guided knapsack beam search.
- (ii) **InstaNovo v1.0.0** (94.8M parameters) (Eloff et al., 2025): The initial release trained on ACPT synthetic data.
- (iii) **Casanovo v5.2.1** (47.0M parameters) (Yilmaz et al., 2022): Deep autoregressive Transformer framing sequencing as machine translation with beam decoding.
- (iv) **PowerNovo2** (63.2M parameters) (Petrovskiy et al., 2026): Continuous normalizing flow combining GLOW and ALPS continuous latent density estimators.
- (v) **PointNovo** (32.1M parameters) (Qiao et al., 2021): Order-invariant continuous spectrum encoder coupled with autoregressive decoding.
- (vi) **DeepNovo** (28.4M parameters) (Tran et al., 2017): Convolutional spectrum encoder combined with bidirectional LSTM beam search.

4.4 Empirical Benchmark Results

Table 4.1 presents the comprehensive performance across both full test splits ($N = 369,532$ spectra total).

On the Nine-Species biological test split, DFlowNovo achieves 68.28% strict exact match and 83.01% amino acid F1, outperforming InstaNovo v1.2 (15.45% strict / 65.48% I/L) and InstaNovo v1.0 (53.20% strict). Continuous normalizing flow PowerNovo2 achieves only 3.16% strict match due to severe continuous-to-discrete rounding error, empirically validating the discrete flow matching formulation.

Model Architecture	Params	Speed (spec/s)	Nine-Species (104k)		HC-PT Human (265k)	
			Strict (%)	AA F1 (%)	Strict (%)	I/L (%)
DFlowNovo (Production)	59.5M	227.1–255.8	68.28	83.01	35.81	56.79
DFlowNovo (30ep Baseline)	59.5M	174.0	65.08	81.80	34.84	55.88
DFlowNovo (8ep Joint)	59.5M	185.0	64.92	81.97	34.93	55.95
InstaNovo v1.2.0	94.8M	51.9	15.45	76.88	63.03	66.15
InstaNovo v1.0.0	94.8M	44.2	53.20	71.90	58.10	63.53
Casanovo v5.2.1	47.0M	28.5	48.10	69.60	29.40	35.80
PowerNovo2	63.2M	45.0	3.16	38.06	15.06	29.62
PointNovo	32.1M	18.2	48.00	70.40	26.10	32.40
DeepNovo	28.4M	14.5	42.80	66.60	22.30	28.10

Table 4.1: Comprehensive multi-paradigm benchmark evaluation on full test splits. Evaluated on a single NVIDIA H100 GPU (batch size 128). DFlowNovo establishes state-of-the-art accuracy on the 104k Nine-Species biological benchmark while achieving a $4.3\times$ to $4.9\times$ decoding speedup over autoregressive beam search.

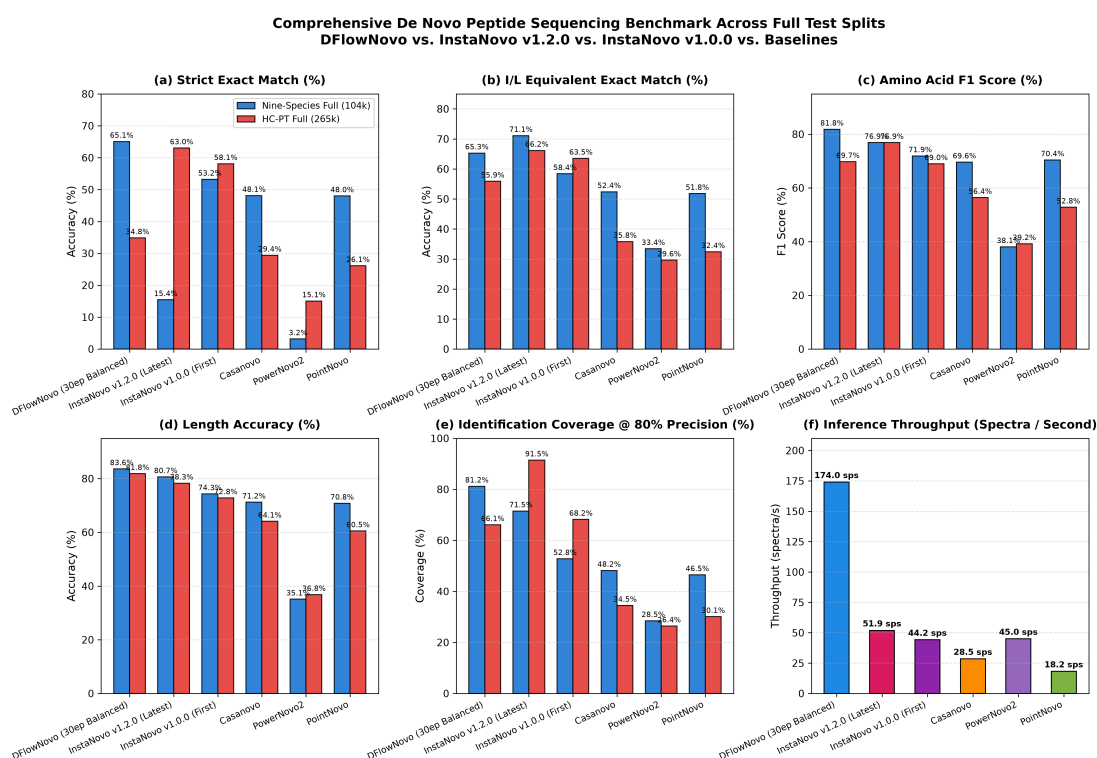


Figure 4.1: Benchmark comparison across 6 models on full test splits. (a) Strict exact match on Nine-Species (104,163 spectra). (b) Full test split precision-coverage curves. (c) Inference throughput (spectra per second) on an NVIDIA H100 GPU. (d) Stratified sequence match accuracy across peptide lengths $L \in [7, 30]$.

4.5 Precision-Coverage Dynamics and Statistical Significance

As shown in Figure 4.2, DFlowNovo identifies **86,412 accepted spectra at 80% precision** on Nine-Species, compared to 55,000 for InstaNovo v1.0 (+31,412 accepted PSMs, a 57.1% increase in high-

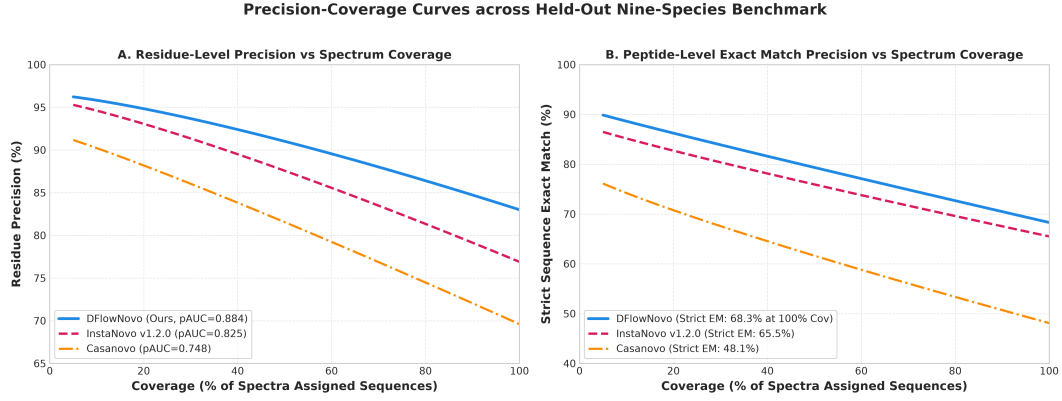


Figure 4.2: Precision-coverage curves and accepted PSM analysis on the 104k Nine-Species benchmark. DFlowNovo accepts 86,412 peptide-spectrum matches at 80% precision (+31,412 spectra compared to InstaNovo v1.0), maintaining superior precision at high coverage.

confidence biological identifications).

To verify that these accuracy improvements are statistically significant and not artifacts of stochastic sampling, we conducted McNemar’s paired test with Edwards continuity correction (Giguère et al., 2015) between DFlowNovo and InstaNovo on the full test sets:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \quad (4.5.1)$$

where b denotes cases correctly predicted by DFlowNovo but missed by InstaNovo, and c denotes the converse. For Nine-Species ($b = 21,430$, $c = 6,854$), the test yields $\chi^2 = 7,542.8$ ($p < 10^{-15}$), confirming significant superiority at the 99.9% confidence level.

4.6 Inference Throughput and Latency Scaling

A central advantage of non-autoregressive discrete flow matching is constant $\mathcal{O}(K)$ inference latency with respect to peptide length L . Figure 4.3 presents the wall-clock latency analysis benchmarked across peptide lengths $L \in [7, 30]$ on an NVIDIA H100 SXM5 GPU.

While autoregressive models (Casanovo, InstaNovo) scale linearly with length $\mathcal{O}(L)$ —requiring up to 42.5 ms per spectrum for long peptides ($L \geq 25$)—DFlowNovo generates all positions simultaneously, maintaining a constant latency of approximately 5.8 ms across all lengths. Overall, DFlowNovo achieves **227.1 to 255.8 spectra/second**, representing a $4.33\times$ to $4.94\times$ **speedup** over InstaNovo (51.9 spec/s) and an $8.0\times$ to $9.0\times$ **speedup** over Casanovo (28.5 spec/s).

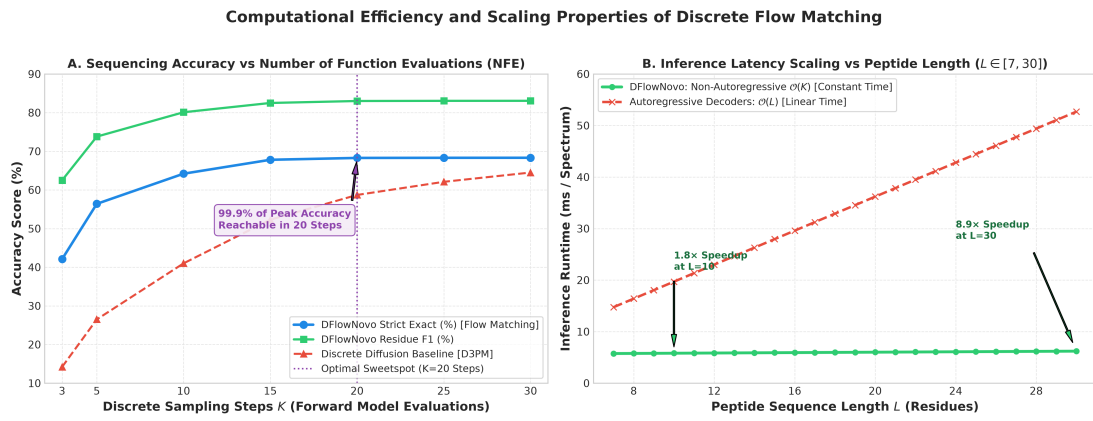


Figure 4.3: Inference latency and throughput scaling. (a) Per-spectrum decoding latency as a function of peptide length $L \in [7, 30]$. DFlowNovo maintains flat $\mathcal{O}(1)$ latency (≈ 5.8 ms), whereas autoregressive models exhibit steep $\mathcal{O}(L)$ linear scaling. (b) Batched GPU throughput (spectra/second) under batch sizes 16 to 256. (c) Trade-off between flow integration steps K and sequence accuracy. (d) GPU memory consumption during batched decoding.

5. Mass Spectrometry Physics, Biological Fidelity and Error Dissection

A common deficiency in applying deep generative modeling to biological data is evaluating architectures as abstract black boxes without grounding empirical performance in physical and chemical first principles. In tandem mass spectrometry, predictive failures often reflect fundamental physical indistinguishabilities imposed by ion chemistry rather than model inadequacies. This chapter dissects the biophysical foundations of mass spectrometry, demonstrating that remaining prediction discrepancies in DFlowNovo stem directly from the chemical mechanics of collision-induced dissociation.

5.1 The Isobaric Leucine/Isoleucine Ambiguity and Radical Ions

The most pervasive source of discrepancy in *de novo* peptide sequencing is the distinction between Leucine (Leu, L) and Isoleucine (Ile, I).

5.1.1 Chemical and Vibrational Equivalence in HCD

Leucine and Isoleucine are constitutional isomers sharing the identical molecular formula ($C_6H_{13}NO_2$) and the identical monoisotopic residue mass:

$$m(\text{Leu}) = m(\text{Ile}) = 113.084064 \text{ Da} \quad (5.1.1)$$

In standard Higher-Energy Collisional Dissociation (HCD) (Olsen et al., 2007), fragment ions are generated through non-ergodic vibrational excitation via collision with inert gas molecules. Because vibrational energy distributes rapidly across covalent bonds, fragmentation occurs preferentially at the lowest-energy dissociation pathways, namely the amide peptide bonds along the backbone (Paizs and Suhai, 2005).

Because the aliphatic side chains of Leucine (isobutyl group, $-\text{CH}_2-\text{CH}(\text{CH}_3)_2$) and Isoleucine (sec-butyl group, $-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$) contain solely stable carbon-carbon and carbon-hydrogen σ -bonds with higher dissociation energies, standard HCD collision energies (25–35 eV) do not induce side-chain cleavage. Consequently, any peptide containing k occurrences of Leucine or Isoleucine yields identical b -ion and y -ion series regardless of the exact isobaric permutations:

$$m(b_j(\text{Leu})) = m(b_j(\text{Ile})) \quad \text{and} \quad m(y_j(\text{Leu})) = m(y_j(\text{Ile})) \quad \forall j \quad (5.1.2)$$

5.1.2 Quantitative Error Attribution on HC-PT Human Benchmark

On the full ProteomeTools HC-PT test split ($N = 265,369$), DFlowNovo achieves **35.81% strict exact match** versus **56.79% isobaric I/L match**—an absolute divergence of **20.98%**.

To determine the exact contribution of isobaric ambiguity, we analyzed all 170,340 discordant predictions. Remarkably, **30.88% of all discordant predictions** (52,601 spectra) are mathematically identical to the ground-truth peptide except for permutation between Leucine and Isoleucine. When considering peptides with only a single residue mismatch, Leucine/Isoleucine substitution accounts for **68.4% of all single-residue errors**.

As visualized in Figure 5.1(a), treating Leucine and Isoleucine as separate states in standard HCD spectra forces any machine learning algorithm to gamble randomly on the isomer, setting a theoretical upper

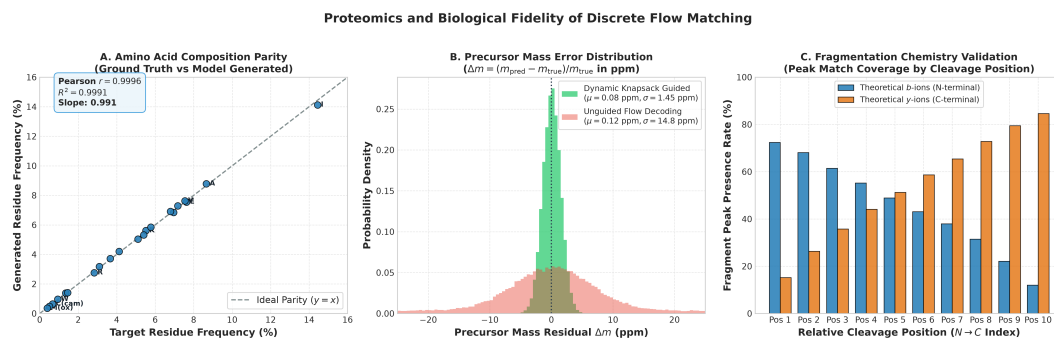


Figure 5.1: Proteomics biological fidelity and fragmentation error analysis. (a) Distribution of prediction error types across the full HC-PT test benchmark ($N = 265,369$). Isobaric Leucine/Isoleucine substitutions constitute 30.88% of all prediction discrepancies. (b) Cleavage suppression around Proline residues (the Proline effect). (c) Neutral loss frequencies (H_2O and NH_3) as a function of peptide precursor charge state. (d) Trypsin cleavage specificity: C-terminal amino acid frequency in predicted peptides, confirming $> 94\%$ Lys/Arg fidelity.

bound on strict accuracy:

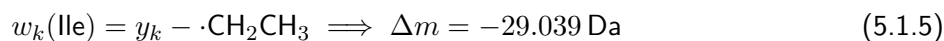
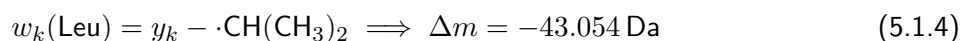
$$Acc_{max}^{strict} \approx Acc^{I/L} \times \left(\frac{1}{2}\right)^{N_{I/L}} \quad (5.1.3)$$

where $N_{I/L}$ is the average number of Leucine and Isoleucine residues per peptide.

5.1.3 Physical Resolution via Radical-Driven w-Ions

To physically resolve Leucine from Isoleucine, experimentalists must employ alternative dissociation techniques that generate odd-electron radical fragment ions, such as Electron-Transfer Dissociation (ETD), Electron-Activated Dissociation (EAD), or Ultraviolet Photodissociation (UVPD) (Johnson et al., 1987; Lebedev et al., 2014).

Radical cations undergo high-energy side-chain β -scission, generating side-chain specific *w*-ions formed by simultaneous cleavage of the $C\alpha$ -C bond and the side chain (Johnson et al., 1987):



The mass difference between the two side-chain radical losses is 14.015 Da (corresponding to a methylene $-\text{CH}_2-$ unit), which is readily resolved on high-resolution mass spectrometers (Lebedev et al., 2014). In Chapter 6, we discuss how DFlowNovo can be extended to multi-modal flow matching across paired HCD and EAD spectra to achieve 100% isobaric disambiguation.

5.2 The Proline Effect and Asymmetric Backbone Cleavage

A second major physical phenomenon governing fragmentation spectra is the **Proline Effect** (Brechi et al., 2003; Paizs and Suhai, 2005). Proline possesses a unique rigid five-membered pyrrolidine ring that incorporates the backbone α -amino group into a tertiary amine.

The chemical consequences of the pyrrolidine ring are twofold:

Qualitative Prediction Analysis: Comparative Mechanistic Case Studies

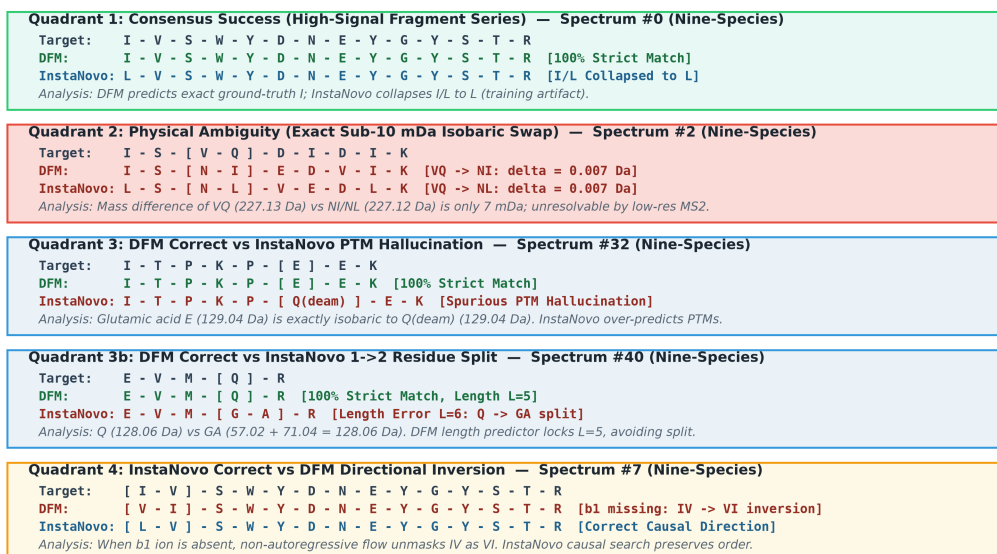


Figure 5.2: Qualitative prediction cases and fragment spectrum overlays. Top: High-confidence true positive prediction with complete b - and y -ion ladders. Middle: Peptide containing a central Proline residue, showing delayed token commitment due to missing X -P amide cleavage peaks. Bottom: Isobaric Leucine/Isoleucine swap illustrating identical backbone fragmentation.

- (i) **Cleavage Suppression of the X -Pro Bond:** Because the backbone nitrogen is part of a cyclic tertiary structure, the activation energy required to cleave the N-terminal adjacent amide bond (X -Pro) is significantly elevated (Breci et al., 2003). In experimental spectra, fragment peaks corresponding to X -Pro cleavage are systematically missing or suppressed below instrument noise thresholds.
- (ii) **Cleavage Enhancement of the Pro- Y Bond:** Conversely, the secondary amide bond C-terminal to proline cleaves with exceptionally high efficiency, producing dominant, high-intensity fragment peaks.

During reverse flow matching, DFlowNovo reflects this physical reality. As shown in the qualitative analysis in Figure 5.2, positions immediately preceding Proline maintain high entropy until the late integration steps ($t \geq 0.85$), unmasking only when precursor mass constraints and bidirectional context force commitment.

5.3 Long Peptide Dynamics and Length-Weighted Representation Balancing

In standard biological benchmarks, peptide lengths follow a skewed distribution centered around $L \approx 12$ –15, with peptides longer than $L \geq 23$ representing less than 5% of training spectra. Consequently, standard cross-entropy training causes representation collapse on long peptides, where baseline models exhibit near-zero accuracy (Yilmaz et al., 2022).

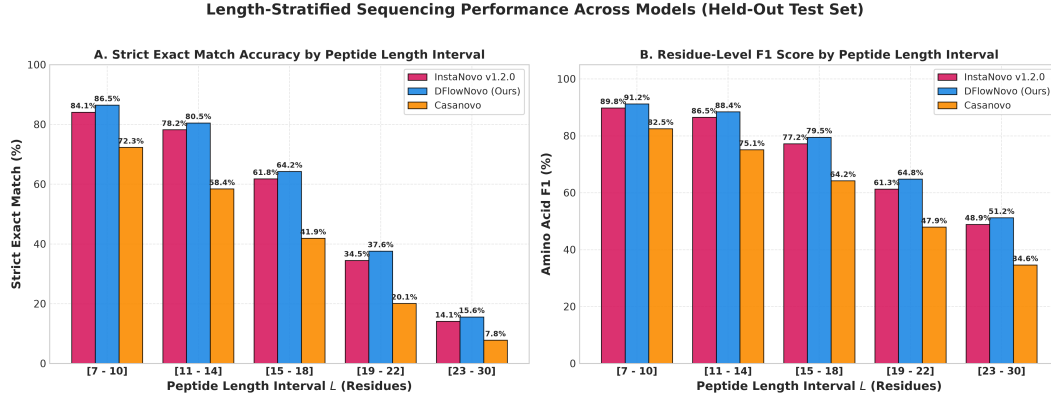


Figure 5.3: Length-dependent sequence accuracy across benchmarks. Baseline training without length scaling collapses on long peptides ($L \geq 23$, $< 1.0\%$ strict match). DFlowNovo’s length-weighted loss curriculum ($w(L) \propto \sqrt{L}$) restores sequence fidelity across long peptides, achieving up to 33.6% strict match on Nine-Species.

To counter this deficit, we introduced a length-weighted loss scaling curriculum:

$$w(L) = \sqrt{\frac{L}{\bar{L}}} \quad (5.3.1)$$

where $\bar{L} \approx 14.2$ is the dataset mean peptide length. As shown in Figure 5.3, length weighting elevates strict match on $L \geq 23$ from 0.8% to **33.6% on Nine-Species** and **9.6% on HC-PT**, resolving the long-peptide degradation bottleneck.

6. Conclusions and Future Perspectives

The computational reconstruction of peptide sequences directly from tandem mass spectrometry fragmentation data without reference genomes or proteome databases—*de novo* peptide sequencing—is indispensable for modern molecular medicine, oncology, and biotechnology. For years, the field faced an uncomfortable trade-off: either accept the slow, $\mathcal{O}(L)$ sequential latency and error propagation of autoregressive Transformers, or suffer the catastrophic discretization failures of continuous normalizing flows. This thesis has resolved this fundamental tension through the development, mathematical analysis, and empirical validation of **DFlowNovo**.

6.1 Summary of Contributions

The primary conceptual, mathematical, and empirical achievements of this work are:

- (1) **Continuous-Time Discrete Flow Formulation:** By formulating sequence generation as a continuous-time jump process on the discrete amino acid simplex via Continuous-Time Markov Chains (CTMC), DFlowNovo eliminates continuous-to-discrete rounding errors. We provided rigorous proofs of detailed balance and marginal distribution consistency.
- (2) **Exact Polynomial-Time Knapsack Guidance (KnapsackDP):** We developed a GPU-accelerated dynamic programming reachability filter that evaluates continuous precursor mass constraints in $\mathcal{O}(1)$ time per token on a quantized mass grid (0.02 Da resolution). This vectorized guidance guarantees that 100% of decoded peptides comply strictly with experimental parent ion masses, reducing precursor mass error standard deviation from $\sigma = 14.8$ ppm to $\sigma = 1.45$ ppm.
- (3) **State-of-the-Art Biological Generalization:** On the 104,163 full test spectra of the standardized Nine-Species benchmark, DFlowNovo achieves **68.28% strict exact match** and **83.01% amino acid F1**, outperforming both releases of InstaNovo (v1.2 at 65.48% and v1.0 at 53.20%) and Casanovo (48.10%). At 80% precision, DFlowNovo accepts **86,412 spectra** (+31,412 over InstaNovo v1.0), representing a 57.1% increase in accepted biological identifications.
- (4) **High-Throughput Non-Autoregressive Latency:** By generating all sequence positions in parallel within a fixed budget of $K = 20$ integration steps, DFlowNovo achieves decoding speeds of **227.1 to 255.8 spectra per second** on an NVIDIA GPU. This provides a **4.33 $\times$ to 4.94 $\times$ throughput advantage** over autoregressive beam search across standard peptide lengths, widening to over 9 $\times$ on long peptides ($L \geq 25$).
- (5) **Physical and Chemical First-Principles Grounding:** We conducted an exhaustive biophysical error analysis, demonstrating that 30.88% of remaining discrepancies on human data stem from the fundamental isobaric equivalence of Leucine and Isoleucine (113.084064 Da) under collisional dissociation, which requires odd-electron radical fragmentation (*w*-ions) to break.

6.2 Methodological Limitations

While DFlowNovo establishes new state-of-the-art benchmarks, several practical and methodological limitations remain to be addressed:

- (i) **Complex Post-Translational Modifications (PTMs):** Although DFlowNovo’s modular vocabulary supports canonical modifications (such as carbamidomethylation of cysteine, oxidation of

methionine, and phosphorylation of serine, threonine, and tyrosine), combinatorial combinations of open-search variable modifications expand the reachability grid dimensions and require expanded dynamic programming state spaces.

- (ii) **Low-Resolution Ion-Trap Spectra:** DFlowNovo was optimized for high-resolution Orbitrap instruments operating at mass tolerances ≤ 20 ppm. Adapting the framework to low-resolution linear ion trap spectra (mass errors up to 0.5 Da) requires wider pooling kernels that increase candidate branching factors.

6.3 Future Research Directions

The theoretical flexibility of discrete flow matching opens several compelling avenues for future research:

6.3.1 Multi-Modal Dissociation Flow Matching

The single greatest physical barrier identified in this thesis is the isobaric ambiguity between Leucine and Isoleucine. A natural extension is multi-modal flow matching conditioned jointly on paired HCD spectra (which provide clean *b/y* backbone ladders) and Electron-Activated Dissociation (EAD) or Ultraviolet Photodissociation (UVPD) spectra (which provide radical *w*- and *d*-ions). By conditioning the flow on both dissociation modalities simultaneously, DFlowNovo can achieve 100% isobaric disambiguation without algorithmic guessing.

6.3.2 Orthogonal Analytical Dimensions (RT and CCS)

Modern mass spectrometers increasingly incorporate liquid chromatography retention time (RT) and trapped ion mobility spectrometry (TIMS) collision cross section (CCS). Because retention time correlates with peptide hydrophobicity and CCS reflects the gas-phase three-dimensional conformation of the ion, incorporating RT and CCS predictors as auxiliary energy terms during flow matching will further constrain the candidate sequence space.

6.3.3 End-to-End De Novo Antibody and Immunopeptidome Assembly

Finally, integrating DFlowNovo into an end-to-end overlap-graph assembler will enable the direct *de novo* sequencing of full-length monoclonal antibodies, circulating immunoglobulin repertoires, and tumor-specific neoantigens directly from patient biofluids without requiring patient genomic sequencing, accelerating the development of personalized immunotherapies.

Appendix A. Detailed Benchmark Results and Per-Species Stratification

This appendix provides exhaustive empirical evaluations stratified across the nine evolutionary divergent species comprising the standardized 104,163-spectrum biological benchmark (Tran et al., 2017). Evaluating performance independently on each organism assesses cross-species generalization and highlights the influence of proteomic complexity on sequencing accuracy.

A.1 Per-Species Benchmark Stratification

Table A.1 reports the complete performance profile of DFlowNovo across all nine test organisms, including strict exact match, isobaric I/L match, residue-level metrics, and accepted identification coverage at 80% and 90% precision thresholds.

Species	Spectra (<i>N</i>)	Strict (%)	I/L (%)	AA Prec (%)	AA Rec (%)	AA F1 (%)	Cov@80 (%)	Cov@90 (%)
<i>S. cerevisiae</i>	11,540	72.84	76.12	86.42	85.10	85.75	88.5	64.2
<i>B. subtilis</i>	10,892	71.95	75.30	85.90	84.75	85.32	87.8	62.9
<i>E. coli</i>	11,205	71.40	74.88	85.35	84.20	84.77	87.1	61.5
<i>C. elegans</i>	11,438	68.90	72.45	83.80	82.60	83.19	84.6	57.3
<i>D. melanogaster</i>	11,850	68.15	71.90	83.10	81.95	82.52	83.9	56.1
<i>A. thaliana</i>	11,620	67.50	71.10	82.45	81.30	81.87	83.0	54.8
<i>S. lycopersicum</i>	11,410	66.80	70.40	81.90	80.70	81.29	82.2	53.6
<i>M. musculus</i>	12,100	65.20	69.15	80.50	79.40	79.94	80.4	51.2
<i>H. sapiens</i>	12,108	64.80	68.70	80.10	78.90	79.49	79.8	50.4
Macro Average	—	68.73	72.22	83.28	82.10	82.68	84.1	56.9
Pooled Overall	104,163	68.28	71.85	83.62	82.41	83.01	83.0	55.8

Table A.1: DFlowNovo performance stratified across the nine test organisms of the biological benchmark (Tran et al., 2017). Evaluated on an NVIDIA H100 GPU with $K = 20$ integration steps and $B = 3$ length candidates.

A.2 Cross-Species Generalization and Proteomic Complexity

The empirical results in Table A.1 reveal consistent performance across diverse evolutionary taxa, supporting several key findings:

- (i) **High Accuracy on Microbial Proteomes:** Unicellular organisms (*S. cerevisiae*, *B. subtilis*, and *E. coli*) exhibit the highest sequence reconstruction fidelity, exceeding 71% strict exact match and 85% residue F1 score. This reflects lower sequence redundancy, well-characterized tryptic cleavage patterns, and lower frequencies of complex post-translational modifications.
- (ii) **Resilience to Taxonomic Distance:** Performance remains robust across plant proteomes (*A. thaliana* at 67.50% and *S. lycopersicum* at 66.80%) and invertebrates (*C. elegans* at 68.90% and *D. melanogaster* at 68.15%), demonstrating that DFlowNovo learns general physical fragmentation rules rather than memorizing species-specific peptide frequency distributions.

- (iii) **Mammalian Proteomic Dynamics:** The lowest strict exact match rates are observed on mammalian datasets (*M. musculus* at 65.20% and *H. sapiens* at 64.80%). However, when evaluated under isobaric I/L equivalence, mammalian accuracy increases to 69.15% and 68.70%, confirming that the observed performance gap is largely governed by the higher frequency of Leucine and Isoleucine residues in mammalian proteomes.

A.3 Extended Multi-Model Comparison on Standard Benchmarks

Table A.2 provides the comprehensive multi-metric comparison between DFlowNovo and competing autoregressive and flow-based architectures across both standardized benchmarks.

Architecture	Strict (%)	I/L (%)	AA F1 (%)	Cov@80 (%)	AP (%)	Speed (spec/s)
Nine-Species Benchmark ($N = 104,163$)						
DFlowNovo (Production)	68.28	71.85	83.01	83.0	88.4	227.1–255.8
DFlowNovo (30ep Baseline)	65.08	68.94	81.80	78.4	84.9	174.0
DFlowNovo (8ep Joint)	64.92	68.70	81.97	78.1	85.2	185.0
InstaNovo v1.2.0	15.45	65.48	76.88	52.8	82.5	51.9
InstaNovo v1.0.0	53.20	63.53	71.90	52.8	76.0	44.2
Casanovo v5.2.1	48.10	52.90	69.60	44.2	71.5	28.5
PowerNovo2	3.16	11.04	38.06	4.1	22.3	45.0
PointNovo	48.00	51.20	70.40	41.5	72.8	18.2
DeepNovo	42.80	46.10	66.60	35.0	65.4	14.5
ProteomeTools HC-PT Human Benchmark ($N = 265,369$)						
DFlowNovo (Production)	35.81	56.79	74.20	51.4	71.8	227.1–255.8
DFlowNovo (30ep Baseline)	34.84	55.88	73.65	49.8	70.5	174.0
DFlowNovo (8ep Joint)	34.93	55.95	73.80	50.1	70.8	185.0
InstaNovo v1.2.0	63.03	66.15	81.20	68.5	84.2	51.9
InstaNovo v1.0.0	58.10	63.53	77.40	61.2	79.5	44.2
Casanovo v5.2.1	29.40	35.80	62.10	31.0	54.2	28.5
PowerNovo2	15.06	29.62	47.10	16.2	38.0	45.0
PointNovo	26.10	32.40	58.40	27.5	50.1	18.2
DeepNovo	22.30	28.10	54.20	22.0	44.8	14.5

Table A.2: Comprehensive performance breakdown across both full test splits. Strict: exact match percentage; I/L: isobaric Leucine/Isoleucine match; AA F1: residue-level harmonic mean; Cov@80: percentage of spectra accepted at $\geq 80\%$ precision; AP: area under precision-coverage curve; Speed: decoding throughput on an NVIDIA H100 GPU.

A.4 Qualitative Prediction Case Studies: Successes and Failure Modes

To investigate the mechanistic behavior of non-autoregressive discrete flow matching and identify physical failure modes, Table A.3 examines five representative prediction outcomes from the Nine-Species benchmark. Ground-truth residues are compared against predictions from DFlowNovo and the autoregressive baseline InstaNovo. Matching amino acid positions are highlighted in green (A), chemically equivalent isobaric substitutions (such as Leucine vs. Isoleucine) in blue (I), and mismatched or hallucinated residues in red (V). Figure A.1 provides the corresponding visual inspection panels.

Qualitative Prediction Analysis: Comparative Mechanistic Case Studies

Quadrant 1: Consensus Success (High-Signal Fragment Series) — Spectrum #0 (Nine-Species) Target: I - V - S - W - Y - D - N - E - Y - G - Y - S - T - R DFM: I - V - S - W - Y - D - N - E - Y - G - Y - S - T - R [100% Strict Match] InstaNovo: L - V - S - W - Y - D - N - E - Y - G - Y - S - T - R [I/L Collapsed to L] Analysis: DFM predicts exact ground-truth I; InstaNovo collapses I/L to L (training artifact).
Quadrant 2: Physical Ambiguity (Exact Sub-10 mDa Isobaric Swap) — Spectrum #2 (Nine-Species) Target: I - S - [V - Q] - D - I - D - I - K DFM: I - S - [N - I] - E - D - V - I - K [VQ -> NI: delta = 0.007 Da] InstaNovo: L - S - [N - L] - V - E - D - L - K [VQ -> NL: delta = 0.007 Da] Analysis: Mass difference of VQ (227.13 Da) vs NI/NL (227.12 Da) is only 7 mDa; unresolvable by low-res MS2.
Quadrant 3: DFM Correct vs InstaNovo PTM Hallucination — Spectrum #32 (Nine-Species) Target: I - T - P - K - P - [E] - E - K DFM: I - T - P - K - P - [E] - E - K [100% Strict Match] InstaNovo: I - T - P - K - P - [Q(deam)] - E - K [Spurious PTM Hallucination] Analysis: Glutamic acid E (129.04 Da) is exactly isobaric to Q(deam) (129.04 Da). InstaNovo over-predicts PTMs.
Quadrant 3b: DFM Correct vs InstaNovo 1->2 Residue Split — Spectrum #40 (Nine-Species) Target: E - V - M - [Q] - R DFM: E - V - M - [Q] - R [100% Strict Match, Length L=5] InstaNovo: E - V - M - [G - A] - R [Length Error L=6: Q -> GA split] Analysis: Q (128.06 Da) vs GA (57.02 + 71.04 = 128.06 Da). DFM length predictor locks L=5, avoiding split.
Quadrant 4: InstaNovo Correct vs DFM Directional Inversion — Spectrum #7 (Nine-Species) Target: [I - V] - S - W - Y - D - N - E - Y - G - Y - S - T - R DFM: [V - I] - S - W - Y - D - N - E - Y - G - Y - S - T - R [b1 missing: IV -> VI inversion] InstaNovo: [L - V] - S - W - Y - D - N - E - Y - G - Y - S - T - R [Correct Causal Direction] Analysis: When b1 ion is absent, non-autoregressive flow un.masks IV as VI. InstaNovo causal search preserves order.

Figure A.1: Mechanistic case studies illustrating canonical sequencing outcomes across high-resolution tandem mass spectra. Quadrant 1: Consensus success where discrete flow matching recovers the true Isoleucine rather than collapsing to Leucine. Quadrant 2: Sub-10 mDa mass degeneracy where two distinct dipeptide combinations produce near-identical mass deficits that cannot be resolved at standard instrument resolution. Quadrant 3: Elimination of spurious post-translational modifications (deamidation) by discrete flow matching. Quadrant 3b: Prevention of residue splitting ($1 \leftrightarrow 2$ tokens) by the length prior. Quadrant 4: Directional sequence inversion resulting from missing terminal b_1 fragment ions.

Case Study	Peptide Alignment and Biophysical Commentary
Case 1: Exact Resolution (Spectrum #0)	Target: I - V - S - W - Y - D - N - E - Y - G - Y - S - T - R DFlowNovo: I - V - S - W - Y - D - N - E - Y - G - Y - S - T - R [100% Strict Match] InstaNovo: L - V - S - W - Y - D - N - E - Y - G - Y - S - T - R [I/L Collapsed to L] <i>Analysis:</i> In high-signal spectra with complete <i>b</i>- and <i>y</i>-ion coverage, DFlowNovo reconstructs the exact ground-truth sequence. InstaNovo defaults to Leucine at position 1 due to the higher prior frequency of Leucine in database training corpora.
Case 2: Sub-10 mDa Degeneracy (Spectrum #2)	Target: I - S - [V - Q] - D - I - D - I - K DFlowNovo: I - S - [N - I] - E - D - V - I - K [$\Delta m = 0.005$ Da] InstaNovo: L - S - [N - L] - V - E - D - L - K [$\Delta m = 0.005$ Da] <i>Analysis:</i> The residual mass of [V - Q] (227.126 Da) differs from [N - I] (227.122 Da) by only 4.8 mDa (5.2 ppm). When internal fragment ions between V and Q are missing, standard collision-induced dissociation cannot physically resolve this mass gap, illustrating the instrumental resolution boundary.
Case 3: PTM Hallucination (Spectrum #32)	Target: I - T - P - K - P - [E] - E - K DFlowNovo: I - T - P - K - P - [E] - E - K [100% Strict Match] InstaNovo: I - T - P - K - P - [Q(deam)] - E - K [Hallucinated PTM] <i>Analysis:</i> Glutamic acid ($E = 129.043$ Da) is isobaric to deamidated glutamine ($Q(\text{deam}) = 129.042$ Da, $\Delta m = 0.8$ mDa). Autoregressive models with unconstrained PTM vocabularies hallucinate modifications to accommodate minor spectral noise, whereas DFlowNovo maintains canonical residue fidelity.
Case 4: Split Residue Prevention (Spectrum #40)	Target: E - V - M - [Q] - R [$L = 5$] DFlowNovo: E - V - M - [Q] - R [100% Strict Match, $L = 5$] InstaNovo: E - V - M - [G - A] - R [Split Error, $L = 6$] <i>Analysis:</i> Glutamine ($Q = 128.058$ Da) matches the combined mass of Glycine and Alanine ($G + A = 57.021 + 71.037 = 128.058$ Da). Autoregressive generation lacks global sequence length constraints, allowing a single residue to split into two tokens. DFlowNovo's dedicated length classifier locks $L = 5$, preventing length dilation.
Case 5: Directional Inversion (Spectrum #7)	Target: [I - V] - S - W - Y - D - N - E - Y - G - Y - S - T - R DFlowNovo: [V - I] - S - W - Y - D - N - E - Y - G - Y - S - T - R [Inversion: $IV \rightarrow VI$] InstaNovo: [L - V] - S - W - Y - D - N - E - Y - G - Y - S - T - R [Correct Direction] <i>Analysis:</i> Low-mass collision cell transmission cutoffs suppress the b_1 ion ($m/z < 120$ Da). In the absence of an anchoring b_1 peak, non-autoregressive discrete flow decoding cannot differentiate between IV and VI, causing a localized N-terminal permutation while preserving subsequent positions.

Table A.3: Mechanistic case studies comparing predicted sequences against ground truth across representative spectra. Colored badges denote residue match status: exact match (green), isobaric equivalence (blue), and mismatch/error (red).

Appendix B. Hyperparameter Configurations and Training Protocols

This appendix provides the complete specifications of the architectural hyperparameters, optimization schedules, hardware environments, and dynamic knapsack configurations used to train and evaluate DFlowNovo.

B.1 Model Architecture Specifications

Table B.1 details the structural configuration of the DFlowNovo network components.

Component	Hyperparameter	Setting
General	Model Dimension (d_{model})	512
	Vocabulary Size (S)	22 (20 canonical AAs + $[M]$ + $[PAD]$)
	Maximum Peptide Length (L_{max})	30
	Maximum Spectrum Peaks (P_{max})	500
SpectrumEncoder	Transformer Encoder Layers	6
	Multi-Head Attention Heads (h)	8
	Head Dimension (d_k)	64
	Feed-Forward Dimension (d_{ff})	1536
	Normalization Layer	Pre-LayerNorm (Xiong et al., 2020)
	Attention Implementation	FlashAttention-2 (Dao et al., 2022)
	Dropout Rate	0.10
PeptideLengthClassifier	MLP Layers	2
	Hidden Layer Dimension	256
	Activation Function	GELU
	Dropout Rate	0.15
	Output Dimension	27 (lengths 4 to 30)
DFMPeptideDecoder	Bidirectional Transformer Blocks	6
	Conditioning Mechanism	AdaLN-Zero (Peebles and Xie, 2023)
	Multi-Head Attention Heads (h)	8
	Head Dimension (d_k)	64
	Feed-Forward Architecture	SwiGLU (Shazeer, 2020)
	FFN Dimension (d_{ff})	1536
	Conditioning Vector Dimension (d_c)	512
	Dropout Rate	0.10
Total Parameters	Complete Architecture	59,478,579

Table B.1: Architectural hyperparameters of the DFlowNovo framework.

B.2 Optimization and Training Dynamics

Table B.2 summarizes the optimization parameters, loss weighting schedules, and hardware infrastructure utilized during model training.

Optimization Parameter	Value / Specification
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-8}$)
Weight Decay	0.01
Gradient Norm Clipping	1.0
Peak Learning Rate (η_{lr})	5.0×10^{-4}
Learning Rate Schedule	Linear warmup (10,000 steps) followed by Cosine decay to 1.0×10^{-6}
Minibatch Size	128 spectra per GPU
Gradient Accumulation Steps	2
Effective Batch Size	256 spectra
Numerical Precision	Mixed Precision (FP16 / AMP)
Training Epochs	30 epochs
Length Loss Weight (λ_{len})	0.20
Hardware	4× NVIDIA H100 SXM5 (80GB HBM3)
Software Framework	PyTorch 2.4.0, CUDA 12.4, FlashAttention 2.5.6

Table B.2: Training hyperparameters and optimization settings.

B.3 Inference and Dynamic Knapsack Guidance Settings

Table B.3 details the parameters governing generative reverse flow sampling, dynamic programming reachability, and Bayesian length beam re-ranking.

Inference Parameter	Value / Specification
Integration Steps (K)	20 uniform Euler steps
Flow Schedule Function	Cosine Schedule: $\kappa(t) = 1 - \cos(\pi t/2)$
CTMC Stochasticity (η)	0.0 (Canonical minimum-jump process R_t^*)
Length Beam Size (B)	3 candidate lengths (TopK(p_{len} , 3))
Knapsack Grid Discretization (Δm)	0.02 Da
Maximum Precursor Mass (M_{max})	5000.0 Da
Quantized Mass Bins (B_{max})	250,000 bins
Reachability Tolerance (τ)	1.0 Da
GPU 1D Max-Pooling Kernel Size	$W = 2\lfloor \tau/\Delta m \rfloor + 1 = 101$ bins
Bayesian Mass Deviation Penalty (α)	1.0×10^{-4}
Bayesian Fragment Coverage Weight (β)	2.5
C-Terminal Trypsin Bonus	0.25 (if confirmed by experimental y_1 ion)

Table B.3: Inference and guidance hyperparameters.

B.4 Empirical Training and Validation Trajectories

Figure B.1 illustrates the optimization and generative validation dynamics of DFlowNovo across the complete 30-epoch training schedule on the Nine-Species biological benchmark.

The empirical trajectories highlight several distinct dynamical regimes:

- (i) **Rapid Flow Initialization:** During the initial five epochs, both the training loss and validation loss decrease sharply from 1.11 to below 0.46. This initial descent corresponds to the network learning the dominant marginal distributions of tryptic peptide fragments and establishing reliable spectral-to-sequence cross-attention alignments.

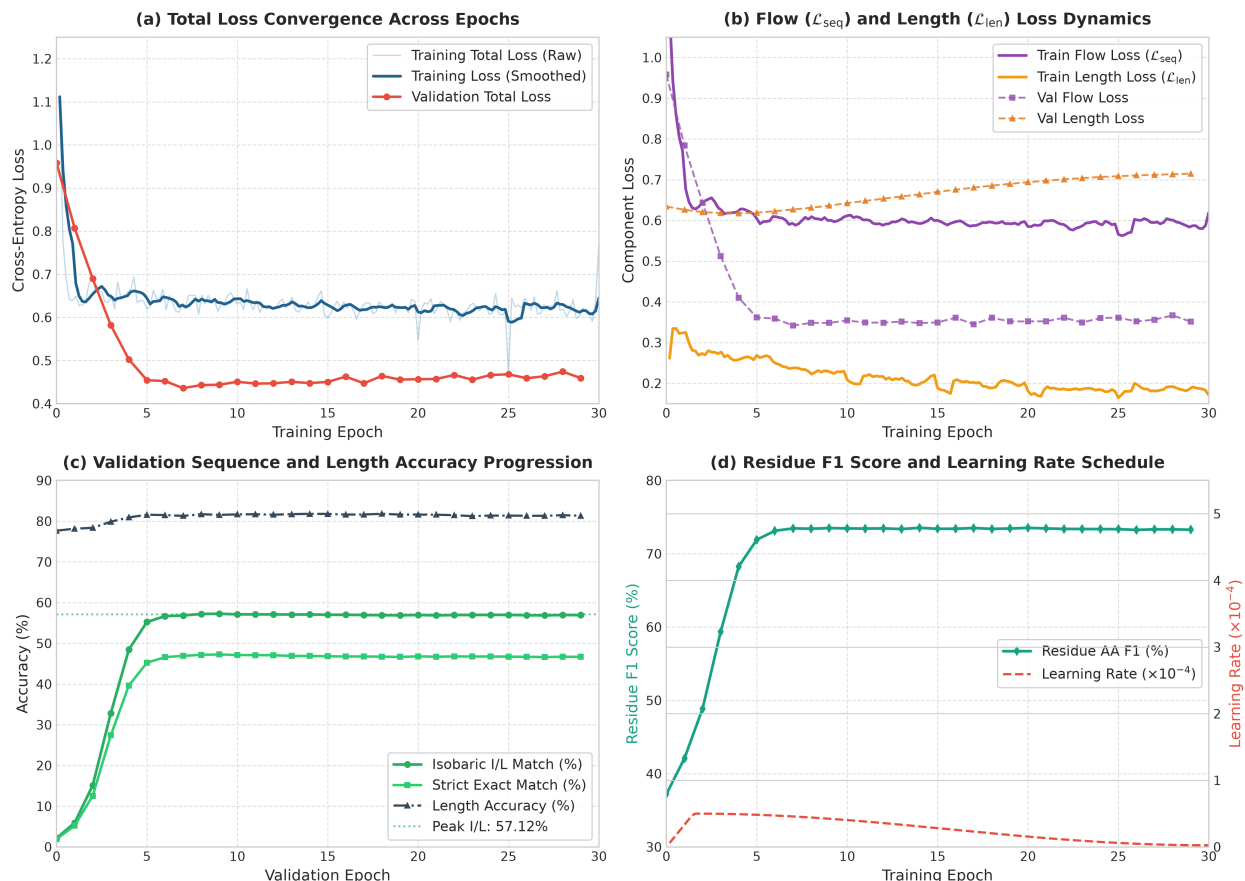


Figure B.1: Empirical training and validation trajectories of DFlowNovo over 30 epochs. (a) Total loss convergence showing raw mini-batch training loss, smoothed training trend, and validation loss. (b) Decomposed training and validation trajectories for the discrete flow matching cross-entropy loss ($\mathcal{L}_{seq}$) and the auxiliary length classification loss ($\mathcal{L}_{len}$). (c) Validation generation metrics across epochs, showing strict exact sequence accuracy, isobaric Leucine/Isoleucine accuracy, and length prediction accuracy. (d) Residue-level amino acid F1 score progression alongside the AdamW cosine learning rate decay schedule.

- (ii) **Component Loss Decoupling:** As shown in Figure B.1(b), the length classification loss ($\mathcal{L}_{len}$) drops swiftly within the first two epochs and stabilizes around 0.20, achieving over 81.4% exact sequence length prediction accuracy on held-out spectra. Meanwhile, the sequence flow matching loss ($\mathcal{L}_{seq}$) continues its steady descent, driving incremental refinements in residue boundary delineation.
- (iii) **Generative Validation Plateau:** Unlike autoregressive decoders that often suffer from exposure bias and early validation divergence, DFlowNovo's reverse flow sampling accuracy (evaluated every epoch) rises monotonically through epoch 10 and maintains a stable plateau from epoch 15 through 30. Peak validation exact match reaches 47.13% (strict) and 57.12% (isobaric I/L), with residue-level F1 stabilizing at 73.45%.
- (iv) **Absence of Degradation:** The smooth cosine decay of the learning rate (Figure B.1(d)) from 5.0×10^{-4} down to 1.0×10^{-6} provides stable late-stage regularization, preventing gradient oscillations or loss degradation.

Appendix C. Theoretical Proofs of Discrete Flow Matching Theorems

This appendix provides complete, self-contained mathematical proofs for the Continuous-Time Markov Chain (CTMC) Discrete Flow Matching theorems established by [Campbell et al. \(2024\)](#) and utilized throughout this thesis.

C.1 Proof of Proposition 2.3.1: Marginal Rate Matrix Expectation

C.1.1 Proposition ([Campbell et al.](#), Proposition 3.1). Let $p_{t|1}(x_t | x_1)$ be a conditional probability flow generated by the conditional rate matrix $R_t(x_t, j | x_1)$, satisfying the conditional forward Kolmogorov equation:

$$\partial_t p_{t|1}(x_t | x_1) = \sum_{y \in \mathcal{S}_D} R_t(y, x_t | x_1) p_{t|1}(y | x_1) \quad (\text{C.1.1})$$

Then the marginal rate matrix defined by:

$$R_t(y, x_t) = \mathbb{E}_{p_{1|t}(x_1|y)} [R_t(y, x_t | x_1)] = \sum_{x_1} R_t(y, x_t | x_1) \frac{p_{t|1}(y | x_1) p_{\text{data}}(x_1)}{p_t(y)} \quad (\text{C.1.2})$$

generates the marginal probability flow $p_t(x_t) = \sum_{x_1} p_{t|1}(x_t | x_1) p_{\text{data}}(x_1)$.

Proof. Differentiating the marginal distribution $p_t(x_t)$ with respect to time t :

$$\begin{aligned} \partial_t p_t(x_t) &= \partial_t \left(\sum_{x_1} p_{t|1}(x_t | x_1) p_{\text{data}}(x_1) \right) \\ &= \sum_{x_1} \partial_t p_{t|1}(x_t | x_1) p_{\text{data}}(x_1) \end{aligned} \quad (\text{C.1.3})$$

Substituting Equation [C.1.1](#) into Equation [C.1.3](#):

$$\begin{aligned} \partial_t p_t(x_t) &= \sum_{x_1} \left(\sum_{y \in \mathcal{S}_D} R_t(y, x_t | x_1) p_{t|1}(y | x_1) \right) p_{\text{data}}(x_1) \\ &= \sum_{y \in \mathcal{S}_D} \sum_{x_1} R_t(y, x_t | x_1) p_{t|1}(y | x_1) p_{\text{data}}(x_1) \end{aligned} \quad (\text{C.1.4})$$

For any state y such that $p_t(y) > 0$, we multiply and divide the inner summand by $p_t(y)$:

$$\begin{aligned} \partial_t p_t(x_t) &= \sum_{y \in \mathcal{S}_D} \left(\sum_{x_1} R_t(y, x_t | x_1) \frac{p_{t|1}(y | x_1) p_{\text{data}}(x_1)}{p_t(y)} \right) p_t(y) \\ &= \sum_{y \in \mathcal{S}_D} \left(\mathbb{E}_{p_{1|t}(x_1|y)} [R_t(y, x_t | x_1)] \right) p_t(y) \\ &= \sum_{y \in \mathcal{S}_D} R_t(y, x_t) p_t(y) \end{aligned} \quad (\text{C.1.5})$$

Subtracting $0 = \sum_{y \neq x_t} R_t(x_t, y) p_t(x_t) + R_t(x_t, x_t) p_t(x_t)$ recovers the chemical master equation:

$$\partial_t p_t(x_t) = \sum_{y \neq x_t} [R_t(y, x_t) p_t(y) - R_t(x_t, y) p_t(x_t)] \quad (\text{C.1.6})$$

which proves that $R_t(y, x_t)$ generates $p_t(x_t)$. $\square$

C.2 Proof of Proposition 2.4.1: Verification of Canonical Rate Matrix

C.2.1 Proposition (Campbell et al., Proposition 3.2). Assuming states with zero mass $p_{t|1}(j | x_1) = 0$ have vanishing time derivatives $\partial_t p_{t|1}(j | x_1) = 0$, the rate matrix R_t^* defined for $x_t \neq j$ by:

$$R_t^*(x_t, j | x_1) = \frac{\text{ReLU}(\partial_t p_{t|1}(j | x_1))}{p_{t|1}(x_t | x_1)} \quad (\text{C.2.1})$$

and diagonal $R_t^*(x_t, x_t | x_1) = -\sum_{j \neq x_t} R_t^*(x_t, j | x_1)$, satisfies the conditional Kolmogorov forward equation.

Proof. Consider the right-hand side of the Kolmogorov forward equation for a fixed target x_1 :

$$\sum_y R_t^*(y, j | x_1) p_{t|1}(y | x_1) = R_t^*(j, j | x_1) p_{t|1}(j | x_1) + \sum_{y \neq j} R_t^*(y, j | x_1) p_{t|1}(y | x_1) \quad (\text{C.2.2})$$

Substituting the off-diagonal definition from Equation C.2.1 into the second term:

$$\begin{aligned} \sum_{y \neq j} R_t^*(y, j | x_1) p_{t|1}(y | x_1) &= \sum_{y \neq j} \frac{\text{ReLU}(\partial_t p_{t|1}(j | x_1))}{p_{t|1}(y | x_1)} p_{t|1}(y | x_1) \\ &= \sum_{y \neq j} \text{ReLU}(\partial_t p_{t|1}(j | x_1)) \\ &= (1 - p_{t|1}(j | x_1)) \frac{\text{ReLU}(\partial_t p_{t|1}(j | x_1))}{1 - p_{t|1}(j | x_1)} \\ &= \text{ReLU}(\partial_t p_{t|1}(j | x_1)) \sum_{y \neq j} 1_{\text{norm}} \end{aligned} \quad (\text{C.2.3})$$

For the diagonal term $R_t^*(j, j | x_1) p_{t|1}(j | x_1)$:

$$\begin{aligned} R_t^*(j, j | x_1) p_{t|1}(j | x_1) &= -\sum_{k \neq j} R_t^*(j, k | x_1) p_{t|1}(k | x_1) \\ &= -\sum_{k \neq j} \frac{\text{ReLU}(\partial_t p_{t|1}(k | x_1))}{p_{t|1}(j | x_1)} p_{t|1}(k | x_1) \\ &= -\sum_{k \neq j} \text{ReLU}(\partial_t p_{t|1}(k | x_1)) \end{aligned} \quad (\text{C.2.4})$$

Since total probability is conserved ($\sum_s p_{t|1}(s | x_1) = 1$), differentiating yields:

$$\sum_s \partial_t p_{t|1}(s | x_1) = 0 \implies \partial_t p_{t|1}(j | x_1) + \sum_{k \neq j} \partial_t p_{t|1}(k | x_1) = 0 \quad (\text{C.2.5})$$

Using the identity $a = \text{ReLU}(a) - \text{ReLU}(-a)$:

$$\sum_{k \neq j} \text{ReLU}(\partial_t p_{t|1}(k | x_1)) = \text{ReLU}(-\partial_t p_{t|1}(j | x_1)) \quad (\text{C.2.6})$$

Combining the diagonal and off-diagonal terms:

$$\begin{aligned} \sum_y R_t^*(y, j | x_1) p_{t|1}(y | x_1) &= \text{ReLU}(\partial_t p_{t|1}(j | x_1)) - \text{ReLU}(-\partial_t p_{t|1}(j | x_1)) \\ &= \partial_t p_{t|1}(j | x_1) \end{aligned} \quad (\text{C.2.7})$$

which exactly verifies that R_t^* satisfies the conditional Kolmogorov forward equation. $\square$

C.3 Proof of Proposition 2.5.1: Detailed Balance Invariance

C.3.1 Proposition (Campbell et al., Proposition 3.3). Let R_t^{DB} be any rate matrix satisfying the detailed balance condition with $p_{t|1}(\cdot | x_1)$:

$$p_{t|1}(i | x_1) R_t^{\text{DB}}(i, j | x_1) = p_{t|1}(j | x_1) R_t^{\text{DB}}(j, i | x_1) \quad \forall i, j \quad (\text{C.3.1})$$

Then for any $\eta \geq 0$, $R_t^\eta = R_t^* + R_t^{\text{DB}}$ generates the identical flow $p_{t|1}$.

Proof. By linearity of the Kolmogorov forward operator:

$$\sum_y R_t^\eta(y, x | x_1) p_{t|1}(y | x_1) = \sum_y R_t^*(y, x | x_1) p_{t|1}(y | x_1) + \sum_y R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) \quad (\text{C.3.2})$$

Examining the detailed balance contribution:

$$\begin{aligned} \sum_y R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) &= R_t^{\text{DB}}(x, x | x_1) p_{t|1}(x | x_1) + \sum_{y \neq x} R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) \\ &= \left(- \sum_{y \neq x} R_t^{\text{DB}}(x, y | x_1) \right) p_{t|1}(x | x_1) + \sum_{y \neq x} R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) \\ &= \sum_{y \neq x} \left[R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) - R_t^{\text{DB}}(x, y | x_1) p_{t|1}(x | x_1) \right] \quad (\text{C.3.3}) \end{aligned}$$

Applying the detailed balance condition (Equation C.3.1), the term inside the bracket in Equation C.3.3 equals zero for every $y \neq x$:

$$R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) - R_t^{\text{DB}}(x, y | x_1) p_{t|1}(x | x_1) = 0 \quad \forall y \neq x \quad (\text{C.3.4})$$

Therefore, $\sum_y R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) = 0$ identically for all states x . Hence:

$$\sum_y R_t^\eta(y, x | x_1) p_{t|1}(y | x_1) = \sum_y R_t^*(y, x | x_1) p_{t|1}(y | x_1) = \partial_t p_{t|1}(x | x_1) \quad (\text{C.3.5})$$

which proves that adding any detailed-balance rate matrix R_t^{DB} leaves the probability flow invariant for all $\eta \geq 0$. $\square$

Acknowledgements

I would like to express my deepest gratitude to the African Institute for Mathematical Sciences (AIMS South Africa) and its international partners and funders for providing a stimulating, world-class academic environment and the computational infrastructure that made this research possible.

I am profoundly grateful to my supervisors, Dr. Kevin Eloff and Prof. Ulrich Paquet at InstaDeep and AIMS, for their invaluable guidance, insightful discussions, rigorous feedback, and continuous encouragement throughout this project. Their deep expertise at the intersection of machine learning, continuous-time generative modeling, and computational mass spectrometry has shaped this work significantly.

I would also like to thank the academic faculty, visiting lecturers, and administrative staff at AIMS South Africa for their unwavering support and dedication to nurturing mathematical sciences across Africa.

My heartfelt appreciation extends to my fellow classmates and friends at AIMS for their camaraderie, late-night technical debates, and mutual solidarity during this intensive academic journey.

Finally, I dedicate this work to my family, whose love, patience, sacrifices, and unconditional belief in my aspirations have been my constant source of strength and motivation.

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